

## Part III

# Solutions to Differential Equations



## Chapter 8

# Taylor Methods for Initial Value Problems

In this chapter we demonstrate how Taylor series expansions combined with the infinite dimensional Radii Polynomial technique introduced in Section 7.6 can be used to rigorously solve initial value problems (IVPs) for systems of ODEs given by polynomial nonlinearities. We begin in Section 8.1 by considering an example of an IVP for a scalar equation. The focus is on explaining the technique with a minimal amount technical complications. As the analysis mimics that of finding a curve of zeros presented in Section 7.7 we recommend the reader review Section 7.7 first. However, there is an additional challenge that needs to be overcome, and therefore we take a short detour to consider a few elementary ideas from complex analysis.

Then in Section 8.4 we discuss the relation between an analytic function and its Taylor coefficients in terms of the Taylor transform. To treat systems of ODEs with arbitrary polynomial nonlinearities we develop a functional analytic setting in Section ?? which leads to general theorems that hold for any IVP in a polynomial vector field (see Section 8.2.1). We illustrate these with an application to the Lorenz system in Section 8.3.

### 8.1 A Scalar IVP

Consider the IVP

$$\dot{x} = x(x - 1), \quad x(t_0) = x_0 \in \mathbb{R}. \quad (8.1)$$

Our goal is as follows. Given an explicit approximate solution  $\bar{x}: U \rightarrow \mathbb{R}$ , prove the existence of  $\tilde{x}: U \rightarrow \mathbb{R}$ , a true solution to (8.1), and provide an explicit bound  $r_0$  such that  $\|\bar{x} - \tilde{x}\|_\infty < r_0$  over  $U$ . We use the Radii Polynomial approach of Section 7.6 and in particular Theorem 7.6.2.

Solutions to differential equations are invariant up to rescaling of time. We exploit this to give ourselves extra numerical freedom. In particular, rescaling time by  $t \rightarrow \frac{t-t_0}{L}$

transforms (8.1) to

$$\dot{x} = f(x) \stackrel{\text{def}}{=} Lx(x-1), \quad x(0) = x_0 \in \mathbb{R}. \quad (8.2)$$

To define a zero finding problem we rewrite (8.2) as

$$\dot{x} - f(x) = \dot{x} - Lx(x-1) = 0, \quad x(0) = x_0 \in \mathbb{R}. \quad (8.3)$$

The first step towards the application of Theorem 7.6.2 is to identify the appropriate Banach spaces  $X$  and  $Y$ . Observe that  $X$  is the space in which the desired zero resides. For the problem being considered here this zero is a solution to (8.2), which has an analytic vector field. By Proposition 19.6.7 solutions  $x$  are analytic functions of time, and therefore, as in Section 7.6 we set  $X = \ell^1_{\nu, \mathbb{N}}$ . However, setting  $Y = \ell^1_{\nu, \mathbb{N}}$  is *not* appropriate. To prepare ourselves for the choice of  $Y$  we first consider some elementary complex analysis.

Given  $r > 0$  and  $z_0 \in \mathbb{C}$ , let  $D_r(z_0) = \{z \in \mathbb{C} \mid |z - z_0| < r\}$ . Set

$$C^\omega(D_r(z_0), \mathbb{C}) \stackrel{\text{def}}{=} \{f: D_r(z_0) \rightarrow \mathbb{C} \mid f \text{ is bounded and analytic}\}.$$

We leave it to the reader to check that with the sup norm,

$$\|f\|_\infty \stackrel{\text{def}}{=} \sup \{|f(z)| \mid z \in D_r(z_0)\},$$

$C^\omega(D_r(z_0), \mathbb{C})$  is a Banach space.

The following proposition is left as an exercise.

**Proposition 8.1.1.** *Let  $h \in C^\omega(D_r(z_0), \mathbb{C})$  with Taylor series  $\sum_{k=0}^\infty a_k(z - z_0)^k$ . If  $s \in (0, r)$ , then*

$$|a_n| \leq \frac{1}{s^n} \|h\|_\infty, \quad \text{for all } n \geq 0.$$

**Proposition 8.1.2.** *Let  $h \in C^\omega(D_r(z_0), \mathbb{C})$  with Taylor series given by  $\sum_{n=0}^\infty a_n(z - z_0)^n$ . If  $0 < \nu < r$ , then  $a = \{a_n\}_{n=0}^\infty \in \ell^1_\nu$ .*

*Proof.* Fix  $\nu \in (0, r)$  and  $s \in (\nu, r)$ . Since  $\overline{D_s(z_0)} \subset D_r(z_0)$ , by Proposition 8.1.1

$$|a_n| \leq \frac{1}{s^n} \|h\|_\infty, \quad \text{for all } n \geq 0.$$

Hence,

$$\|a\|_{1, \nu} = \sum_{n=0}^\infty |a_n| \nu^n = \sum_{n=0}^\infty |a_n| s^n \left(\frac{\nu}{s}\right)^n \leq \sum_{n=0}^\infty \|h\|_\infty \left(\frac{\nu}{s}\right)^n = \frac{s}{s - \nu} \|h\|_\infty < \infty. \quad \square$$

We can also work in the opposite direction.

**Proposition 8.1.3.** *Let  $a = \{a_n\}_{n=0}^\infty \in \ell_\nu^1$  and define*

$$h(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n. \quad (8.4)$$

*Then  $h \in C^\omega(D_\nu(z_0), \mathbb{C})$  and  $\|h\|_\infty \leq \|a\|_{1,\nu}$ .*

*Proof.* The estimate

$$\begin{aligned} \|a\|_{1,\nu} &= \sum_{n=0}^{\infty} |a_n| \nu^n \\ &\geq \sup_{|z-z_0|<\nu} \sum_{n=0}^{\infty} |a_n| |z - z_0|^n \\ &\geq \sup_{|z-z_0|<\nu} \left| \sum_{n=0}^{\infty} a_n (z - z_0)^n \right| \\ &= \sup_{|z-z_0|<\nu} |h(z)| \\ &= \|h\|_\infty \end{aligned}$$

shows that the sum in (8.46) converges and hence that  $h \in C^\omega(D_\nu(z_0), \mathbb{C})$ .  $\square$

Returning to IVP we know that the desired solution  $x$  is analytic, i.e.,

$$x(t) = \sum_{n=0}^{\infty} a_n t^n. \quad (8.5)$$

If we assume  $x \in D_r(0)$ , then by Proposition 8.1.2  $a = \{a_n\}_{n=0}^\infty \in \ell_\nu^1$  for any  $\nu \in (0, r)$ . Thus, we have justified our choice of  $X = \ell_\nu^1$ .

If  $x$  is a solution to the initial value problem, then

$$x(0) = a_0 = x_0$$

and more generally substituting (8.5) into (8.3) produces

$$\sum_{n=0}^{\infty} (n+1) a_{n+1} t^n + L \sum_{n=0}^{\infty} a_n t^n - L \sum_{n=0}^{\infty} (a * a)_n t^n = 0,$$

where

$$(a * a)_n = \sum_{j=0}^n a_{n-j} a_j$$

is the Cauchy product. Matching like powers in  $t$  gives an infinite sequence of equations,

$$(n+1)a_{n+1} + La_n - L(a * a)_n = 0, \quad n \geq 0. \quad (8.6)$$

Define  $F: \ell_\nu^1 \rightarrow S_{\mathbb{N}}(\mathbb{R})$  by

$$F_n(a) \stackrel{\text{def}}{=} \begin{cases} a_0 - x_0 & \text{for } n = 0 \\ na_n + L(a_{n-1} - (a * a)_{n-1}) & \text{for } n \geq 1. \end{cases} \quad (8.7)$$

If  $F(a) = 0$ , then  $x: (-\nu, \nu) \rightarrow \mathbb{R}$  as defined by (8.5) is solution to the initial value problem (8.3).

**Lemma 8.1.4.** *Let  $\nu > 0$ . If  $a \in \ell_\nu^1$  satisfies  $F(a) = 0$ , then  $x(t)$  defined by (8.5) solves the initial value problem (8.3) for  $t \in (-\nu, \nu)$ .*

*Proof.* Since  $a \in \ell_\nu^1$  the power series in (8.5) has radius of convergence at least  $\nu$ . Let  $x(t)$  be defined by (8.5) for  $t \in (-\nu, \nu)$ , then  $x(0) = a_0 = x_0$  since  $F(a)_0 = 0$ . Therefore,  $x(t)$  satisfies the initial data.

In the interior of the interval of convergence, differentiation of power series can be done term by term, hence  $\dot{x}(t) = \sum_{n=1}^{\infty} na_n t^{n-1}$  for all  $t \in (-\nu, \nu)$ . Since  $\ell_\nu^1$  is a Banach algebra under the Cauchy product  $*$ , we infer that  $Lx(x-1) = \sum_{n=0}^{\infty} L(a_{n-1} - (a * a)_{n-1}) t^n$  for all  $t \in (-\nu, \nu)$  and  $n \geq 1$ . It now follows from  $F(a)_n = 0$  that  $x'(t) = f(x(t))$  for  $t \in (-\nu, \nu)$ , hence  $x(t)$  solves the initial value problem on this time interval.  $\square$

It is important to note that we have (for the moment) defined  $S_{\mathbb{N}}(\mathbb{R})$  to be the range for  $F$ . This is clearly not desirable as  $S_{\mathbb{N}}(\mathbb{R})$  is not a Banach space. To understand the issue observe that

$$\frac{dx}{dt}(t) = \sum_{n=0}^{\infty} (n+1)a_{n+1}t^n,$$

but the assumption that  $a = \{a_n\}_{n=0}^{\infty} \in \ell_\nu^1$  does not imply that  $\{(n+1)a_{n+1}\}_{n=0}^{\infty} \in \ell_\nu^1$ . The following proposition, the proof of which involves a direct computation, allows us to resolve the issue of choosing an appropriate range for  $F$ .

**Proposition 8.1.5.** *For fixed  $\nu > 0$  let  $\omega = \left\{ \frac{\nu^{n+1}}{n+1} \right\}_{n=0}^{\infty}$  be a sequence of weights. If  $a = \{a_n\}_{n=0}^{\infty} \in \ell_\nu^1$ , then  $a \in \ell_\omega^1$  and  $b = \{(n+1)a_{n+1}\}_{n=0}^{\infty} \in \ell_\omega^1$ .*

As a consequence of Proposition 10.1.1 the image of  $F$  is contained in  $\ell_\omega^1$  and thus, we define  $Y = \ell_\omega^1$  where  $\omega = \left\{ \frac{\nu^{n+1}}{n+1} \right\}_{n=0}^{\infty}$ .

The second step towards the application of Theorem 7.6.2 is the identification of the approximate solution  $\bar{a} \in \ell_\nu^1$ .

For any choice of  $N$  the first  $N+1$  functions  $\{F_n \mid n = 0, \dots, N\}$  are determined entirely by the first  $N+1$  variables  $\{a_n \mid n = 0, \dots, N\}$ . Therefore, we can define  $F^{(N)}: \mathbb{R}^{N+1} \rightarrow \mathbb{R}^{N+1}$  by

$$F_n^{(N)}(a) \stackrel{\text{def}}{=} \begin{cases} a_0 - x_0 & \text{for } n = 0 \\ na_n + L(a_{n-1} - (a * a)_{n-1}) & \text{for } 1 \leq n \leq N, \end{cases} \quad (8.8)$$

and attempt to solve a truncated problem

$$F_n^{(N)}(a_0, \dots, a_N) = 0, \quad n = 0, \dots, N. \quad (8.9)$$

This is a finite dimensional zero finding problem which can be solved numerically. Let  $\bar{a}^{(N)} = (\bar{a}_0, \bar{a}_1, \dots, \bar{a}_N)$  denote such a numerically determined zero. Define

$$\bar{a} := (\bar{a}_0, \bar{a}_1, \dots, \bar{a}_{N-1}, \bar{a}_N, 0, 0, 0, \dots) \in \ell_\nu^1.$$

and observe that

$$\bar{x}^{(N)}(t) := \sum_{n=0}^{\infty} \bar{a}_n t^n = \sum_{n=0}^N \bar{a}_n t^n$$

is an approximate solution to (8.3).

The third step towards the application of Theorem 7.6.2 is the choice of a linear operator  $A^\dagger: X \rightarrow Y$  that approximates  $DF(\bar{a})$ .

We start by analyzing the entries of  $(DF(a))_{n,j}$ ,  $n, j \geq 0$ . For  $n = 0$  we have

$$(DF(a))_{0,j} = -\delta_{0,j},$$

where  $\delta_{n,j}$  denotes the Kronecker delta. For  $n \geq 1$ ,

$$\begin{aligned} (DF(a))_{n,j} &= \frac{\partial F_n}{\partial a_j}(a) \\ &= \frac{\partial}{\partial a_j} (na_n + La_{n-1} - L(a * a)_{n-1}) \\ &= \begin{cases} L(\delta_{n-1,j} - 2a_{n-1-j}) & \text{for } j < n, \\ n & \text{for } j = n, \\ 0 & \text{for } j > n. \end{cases} \end{aligned}$$

In particular, for  $j > n$  the derivative  $\frac{\partial F_n}{\partial a_j}$  vanishes, since  $F_n$  only depends on the variables  $a_0, \dots, a_n$ . Hence the derivative is lower triangular and conceptually we may view the operator  $DF(\bar{a})$  as a lower triangular infinite-dimensional matrix

$$DF(\bar{a}) = \begin{bmatrix} DF^{(N)}(\bar{a}) & \mathbf{0} & \mathbf{0} & \dots \\ * & N+1 & 0 & \dots \\ * & * & N+2 & \dots \\ \vdots & \vdots & \ddots & \ddots \end{bmatrix} \quad (8.10)$$

Notice that while the off diagonal terms are bounded, the diagonal of  $DF(\bar{a})$  is unbounded and goes to infinity. This implies that asymptotically  $DF(\bar{a})$  is diagonally dominant and suggests approximating it by a matrix of the form

$$A^\dagger = \begin{bmatrix} DF^{(N)}(\bar{a}) & 0 \\ 0 & \Lambda \end{bmatrix}$$

where  $\Lambda$  is an infinite diagonal matrix with diagonal entries  $n$ , starting at  $N + 1$ . More precisely, let  $A^\dagger$  be the linear operator whose action on a vector  $h = \{h_n\}_{n \in \mathbb{N}} \in \ell_\nu^1$  is given by

$$(A^\dagger h)_n \stackrel{\text{def}}{=} \begin{cases} [DF^{(N)}(\bar{a})h^{(N)}]_n, & 0 \leq n \leq N \\ nh_n, & n \geq N + 1, \end{cases}$$

where  $DF^{(N)}(\bar{a})$  is the Jacobian matrix of  $F^{(N)}$  at the numerical approximation  $\bar{a}$ . Here  $h^{(N)} = (h_0, h_1, \dots, h_N) \in \mathbb{R}^{N+1}$  is obtained by projecting  $h$  onto its first  $(N + 1)$  coordinates. That  $A^\dagger: \ell_\nu^1 \rightarrow \ell_\omega^1$  follows from Proposition 10.1.1.

The fourth step towards the application of Theorem 7.6.2 is the choice of a linear operator  $A: Y \rightarrow X$  that approximates  $(A^\dagger)^{-1}$ . Since the matrix  $DF^{(N)}(\bar{a}) \in M_{N+1}(\mathbb{R})$  is lower triangular with positive elements on the diagonal, it is invertible. Let  $A^{(N)}$  be an approximate inverse of  $DF^{(N)}(\bar{a})$  (if  $N$  is not too large then we can choose  $A^{(N)}$  to be a numerical inverse of  $DF^{(N)}(\bar{a})$ ). Conceptually, we choose

$$A = \begin{bmatrix} A^{(N)} & 0 \\ 0 & \Lambda^{-1} \end{bmatrix}$$

and, more precisely, we define  $A$  by setting

$$(Ah)_n \stackrel{\text{def}}{=} \begin{cases} [A^{(N)}h^{(N)}]_n & 0 \leq n \leq N \\ \frac{1}{n}h_n & n \geq N + 1. \end{cases} \quad (8.11)$$

**Proposition 8.1.6.**  $A \in B(Y, X)$

*Proof.* Observe that

$$\begin{aligned} \|Ah\|_{1,\nu} &= \sum_{n=0}^N |Ah|_n \nu^n + \sum_{n=N+1}^{\infty} \frac{|h_n|}{n} \nu^n \\ &= K_h + \frac{1}{\nu} \sum_{n=N+1}^{\infty} |h_n| \frac{\nu^{n+1}}{n+1} \frac{n+1}{n} \\ &\leq K_h + \frac{2}{\nu} \|h\|_{1,\omega}, \end{aligned}$$

which proves that  $A: Y \rightarrow X$ . To see that  $A \in B(Y, X)$  observe that there exists a uniform bound  $K$  such that if  $\|h\|_{1,\omega} = 1$ , then  $K_h \leq K$ .  $\square$



We leave it to the reader to check that  $A \in B(X, X)$ .

A direct application of Proposition 7.6.5 implies that if we identify a value  $r_0$  at which the radii polynomial  $p(r_0) < 0$ , then  $A$  is injective.

To apply Theorem 7.6.2 we need the  $Y_0$ ,  $Z_0$ ,  $Z_1$ , and  $Z_2$  bounds, which are collected in the following result.

**Theorem 8.1.7.** *Fix  $L$ ,  $N$ , and  $\nu$ . Let  $F^{(N)}: \mathbb{R}^{N+1} \rightarrow \mathbb{R}^{N+1}$  be defined by (8.8). Let  $\bar{a}^{(N)} = (\bar{a}_0, \dots, \bar{a}_N)$  be an approximate solution to  $F^{(N)}(a) = 0$  and let  $\bar{a}$  be given by*

$$\bar{a} := (\bar{a}_0, \dots, \bar{a}_N, 0, 0, 0, \dots) \in \ell_\nu^1.$$

Let  $A^{(N)}$  be a numerical inverse of  $DF^{(N)}(\bar{a})$ . Set

$$\begin{aligned} Y_0 &\stackrel{\text{def}}{=} \sum_{n=0}^N \left| \left( A^{(N)} F^{(N)}(\bar{a}) \right)_n \right| \nu^n + L\nu \sum_{n=N}^{2N} \frac{1}{n+1} |(\bar{a} - \bar{a} * \bar{a})_n| \nu^n, \\ Z_0 &\stackrel{\text{def}}{=} \max_{0 \leq j \leq N} \frac{1}{\nu^j} \sum_{0 \leq i \leq N} \left| \left( I - A^{(N)} DF^{(N)}(\bar{a}) \right)_{i,j} \right| \nu^i, \\ Z_1 &\stackrel{\text{def}}{=} \frac{L\nu}{N+1} \|\bar{g}\|_{1,\nu}, \\ Z_2 &\stackrel{\text{def}}{=} 2L\nu \|A\|_{B(\ell_\nu^1)}, \end{aligned}$$

where  $\bar{g} = \{\delta_{n,0} - 2\bar{a}_n\}_{n=0}^\infty$ .

Then,  $Y_0$ ,  $Z_0$ ,  $Z_1$ , and  $Z_2$  satisfy the hypotheses of Theorem 7.6.2.

We prove Theorem 8.1.7 via a series of lemmas

**Lemma 8.1.8.** *Under the conditions of Theorem 8.1.7*

$$\|AF(\bar{a})\|_{1,\nu} \leq Y_0.$$

*Proof.* Observe that

$$\begin{aligned} \|AF(\bar{a})\|_{1,\nu} &= \sum_{n=0}^N \left| \left( A^{(N)} F^{(N)}(\bar{a}) \right)_n \right| \nu^n + \sum_{n=N+1}^\infty \left| \frac{1}{n} (n\bar{a}_n - L(\bar{a}_{n-1} - (\bar{a} * \bar{a})_{n-1})) \right| \nu^n \\ &= \sum_{n=0}^N \left| \left( A^{(N)} F^{(N)}(\bar{a}) \right)_n \right| \nu^n + L \sum_{n=N+1}^{2N+1} \frac{1}{n} |\bar{a}_{n-1} - (\bar{a} * \bar{a})_{n-1}| \nu^n \\ &= \sum_{n=0}^N \left| \left( A^{(N)} F^{(N)}(\bar{a}) \right)_n \right| \nu^n + L\nu \sum_{n=N}^{2N} \frac{1}{n+1} |(\bar{a} - \bar{a} * \bar{a})_n| \nu^n. \quad \square \end{aligned}$$

**Lemma 8.1.9.** *Under the conditions of Theorem 8.1.7*

$$\|I - AA^\dagger\|_{B(\ell_\nu^1)} \leq Z_0.$$

*Proof.* Define  $B = I - AA^\dagger$ . Observe that if  $n > N$  or  $j > N$ , then

$$B_{n,j} = \left( I - AA^\dagger \right)_{n,j} = \delta_{n,j} - \frac{1}{n} n \delta_{n,j} = 0.$$

The infinite dimensional operator  $B$  can thus be represented by the finite matrix

$$B^{(N)} = I - A^{(N)} DF^{(N)}(\bar{a}).$$

Its matrix norm, induced by  $\ell_\nu^1$ , is given by

$$\|B^{(N)}\|_{\nu,N} := \max_{0 \leq j \leq N} \frac{1}{\nu^j} \sum_{0 \leq i \leq N} |B_{i,j}^{(N)}| \nu^i \quad (8.12)$$

(see Exercise 2.7.2). Finally, observe that

$$\|I - AA^\dagger\|_{B(\ell_\nu^1)} \leq \|B^{(N)}\|_{\nu,N} = Z_0. \quad \square$$

**Lemma 8.1.10.** *Under the conditions of Theorem 8.1.7*

$$\|A[DF(\bar{a}) - A^\dagger]\|_{B(X)} \leq Z_1.$$

*Proof.* Given  $h \in \ell_\nu^1$  with  $\|h\|_{1,\nu} \leq 1$ , set

$$z \stackrel{\text{def}}{=} [DF(\bar{a}) - A^\dagger]h.$$

Note that

$$z_n = \begin{cases} 0 & n = 0, \dots, N \\ -L(h_{n-1} - 2(\bar{a} * h)_{n-1}) & n > N, \end{cases}$$

and therefore,  $z^{(N)} = (z_0, z_1, \dots, z_N) = 0 \in \mathbb{R}^{N+1}$ . We can express the non-vanishing terms more compactly by introducing

$$\bar{g}_n \stackrel{\text{def}}{=} \delta_{n,0} - 2\bar{a}_n,$$

which vanishes again for  $n > N$ . With this notation we have

$$h_{n-1} - 2(\bar{a} * h)_{n-1} = (\bar{g} * h)_{n-1}.$$

Hence,

$$\begin{aligned} \|Az\|_{1,\nu} &= \sum_{n=0}^N \left| (A^{(N)} z^{(N)})_n \right| \nu^n + \sum_{n \geq N+1} \frac{1}{n} |z_n| \nu^n \\ &\leq \frac{L\nu}{N+1} \sum_{n \geq N+1} |(\bar{g} * h)_{n-1}| \nu^{n-1} \\ &\leq \frac{L\nu}{N+1} \|\bar{g} * h\|_{1,\nu} \\ &\leq \frac{L\nu}{N+1} \|\bar{g}\|_{1,\nu}. \end{aligned}$$

Thus we obtain the desired inequality for

$$Z_1 \stackrel{\text{def}}{=} \frac{L\nu}{N+1} \|\bar{g}\|_{1,\nu}. \quad \square$$

**Lemma 8.1.11.** *Under the conditions of Theorem 8.1.7*

$$\|A[DF(c) - DF(\bar{a})]\|_{B(\ell_\nu^1)} \leq Z_2 r, \quad \text{for all } c \in \overline{B_r(\bar{a})}.$$

*Proof.* Let  $h \in \ell_\nu^1$  with  $\|h\|_{1,\nu} \leq 1$  and let  $c \in \overline{B_r(\bar{a})}$ . Then, for  $n \geq 1$

$$\begin{aligned} ([DF(c) - DF(\bar{a})]h)_n &= nh_n + Lh_{n-1} - 2L(c * h)_{n-1} - (nh_n + Lh_{n-1} - 2L(\bar{a} * h)_{n-1}) \\ &= -2L((c - \bar{a}) * h)_{n-1} \end{aligned}$$

and therefore

$$\begin{aligned} \|[DF(c) - DF(\bar{a})]h\|_{1,\nu} &= 2L \sum_{n \geq 1} |((c - \bar{a}) * h)_{n-1}| \nu^n \\ &= 2L\nu \sum_{n \geq 1} |((c - \bar{a}) * h)_{n-1}| \nu^{n-1} \\ &= 2L\nu \|(c - \bar{a}) * h\|_{1,\nu} \\ &\leq 2L\nu r. \end{aligned}$$

We conclude that for all  $c \in \overline{B_r(\bar{a})}$ ,

$$\begin{aligned} \|A[DF(c) - DF(\bar{a})]\|_{B(\ell_\nu^1)} &= \sup_{\|h\|_{1,\nu} \leq 1} \|A[DF(c) - DF(\bar{a})]h\|_{1,\nu} \\ &\leq \|A\|_{B(\ell_\nu^1)} \|[DF(c) - DF(\bar{a})]h\|_{1,\nu} \\ &\leq \|A\|_{B(\ell_\nu^1)} 2L\nu r. \end{aligned}$$

where

$$\|A\|_{B(\ell_\nu^1)} = \max \left\{ \|A^{(N)}\|_{1,\nu}, \frac{1}{N+1} \right\}$$

(see Exercise 2.7.2). □

We are now in a position to apply Theorem 7.6.2 to solve the IVP (8.1). Consider the initial condition  $x(0) = 1/2$ . Choose  $L = 1$ ,  $N = 27$  and  $\nu = 1$ . Solving for the approximate solution to  $F^{(27)}(a^{(27)}) = 0$  we obtain

$$\bar{a}^{(27)} =$$

Applying Theorem 8.1.7 gives rise to numerical values for  $Y_0$ ,  $Z_0$ ,  $Z_1$ , and  $Z_2$ . For these values  $p(2.72 \times 10^{-15}) < 0$  and hence we set  $r_0 = 2.72 \times 10^{-15}$ .

Therefore, by Theorem 7.6.2 there exists a unique solution  $\tilde{a} \in \overline{B_{r_0}(\bar{a})}$  such that  $F(\tilde{a}) = 0$ . When comparing the true solution  $\tilde{x}(t) \stackrel{\text{def}}{=} \sum_{n=0}^{\infty} \tilde{a}_n t^n$  to the numerically computed approximation  $\bar{x}^{(27)}(t) \stackrel{\text{def}}{=} \sum_{n=0}^{27} \bar{a}_n t^n$ , we obtain the error bound

$$\left\| \tilde{x}(t) - \bar{x}^{(27)}(t) \right\|_{\infty} \leq r_0 = 2.72 \times 10^{-15}.$$

Figure 8.1 provides graphs of approximate solutions to the IVP (8.2) given the choices  $L = 3$ ,  $N = 300$  and  $\nu = 1$ .

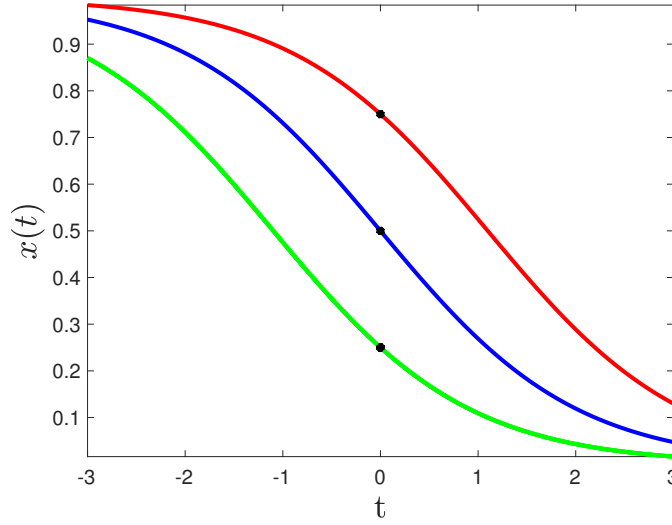


Figure 8.1: Three solutions of  $\dot{x} = Lx(x - 1)$  with  $L = 3$  and  $\nu = 1$ . The green curve has  $x(0) = 1/4$ , the blue curve  $x(0) = 1/2$ , and the red curve  $x(0) = 3/4$ . Each is computed with  $N = 300$  Taylor coefficients. We are able to obtain validated error bounds of  $r = 1.2 \times 10^{-13}$ ,  $r = 6.97 \times 10^{-6}$ , and  $2.3 \times 10^{-13}$  for the green, blue, and red curves respectively.

## 8.2 IVP for Systems of ODEs

In this section we consider the general IVP

$$\dot{x} = f(x), \quad x(0) = x_0 \in \mathbb{R}^L, \quad (8.13)$$

where  $f = (f_1, \dots, f_L)$  is a polynomial nonlinearity. By Proposition 19.6.7, the fact that we are working with polynomial nonlinearities implies that the solutions to (8.13) will be

analytic. More precisely, if  $x: U \rightarrow \mathbb{R}^L$  is a solution, then  $x_l: U \rightarrow \mathbb{R}$  is an analytic function for each  $l = 1, \dots, L$ . This, in turn, implies that

$$x_l(t) = \sum_{n=0}^{\infty} a_{l,n} t^n, \quad l = 1, \dots, L, \quad (8.14)$$

where  $a_l := \{a_{l,n}\}_{n=0}^{\infty} \in \ell_{\nu}^1$ , for some  $\nu > 0$ . We set

$$a = (a_1, \dots, a_L) \in (\ell_{\nu}^1)^L.$$

As in the previous section, given  $\bar{x}$  an approximate solution to the IVP (8.13) our goal is to use Theorem 7.6.2 to prove the existence of a true solution  $\tilde{x}$  and a bound  $r_0$  such that  $\|\bar{x} - \tilde{x}\|_{\infty} < r_0$ . The zero finding problem is identify  $x: (-\nu, \nu) \rightarrow \mathbb{R}^L$  such that

$$\dot{x} - f(x) = 0$$

subject to the constraint that  $x(0) = x_0$ . The essential functional analytic details are dealt with in Section 8.1. However, working with systems of equations leads to more complicated bookkeeping, and therefore, the major effort of this section is to develop appropriate notation to deal with vector valued sequences.

The first step in the application of Theorem 7.6.2 is to identify the Banach spaces  $X$  and  $Y$ . As suggested above we set  $X = (\ell_{\nu}^1)^L$ . Therefore, as is discussed at the end of Section 7.4, given  $a \in (\ell_{\nu}^1)^L$  we express the action of  $f$  on elements in  $(\ell_{\nu}^1)^L$  using the Cauchy product.

Our goal is to simultaneously solve the  $L$  infinite systems of equations

$$\begin{aligned} \dot{x}_l(t) - f_l(x(t)) &= \sum_{n=1}^{\infty} n a_{l,n} t^{n-1} - \sum_{n=0}^{\infty} \phi_l(a)_n t^n \\ &= \sum_{n=1}^{\infty} (n a_{l,n} - \phi_l(a)_{n-1}) t^{n-1} \\ &= 0 \end{aligned}$$

obtained by solving for the coefficients of the power series. For  $\nu' \in (0, \nu)$  define  $F: (\ell_{\nu}^1)^L \rightarrow (\ell_{\nu'}^1)^L$  component-wise by

$$(F_l(a))_n \stackrel{\text{def}}{=} \begin{cases} a_{l,0} - x_{0,l} & \text{if } n = 0, \\ n a_{l,n} - \phi_l(a)_{n-1} & \text{if } n \geq 1. \end{cases} \quad (8.15)$$

The first term is imposed by the initial condition  $x_l(0) = a_{l,0} = x_{0,l}$ . Applying the proof of Lemma 8.1.4 componentwise demonstrates that if  $F(a) = 0$ , then (8.14) is a solution to the IVP 8.13. It is left to reader to check that by Proposition 10.1.1  $(\ell_{\nu'}^1)^L$  is an appropriate choice for the range of  $F$ . Thus,  $Y = (\ell_{\nu'}^1)^L$ .

The second step in the application of Theorem 7.6.2 is the identification of an approximate solution  $\bar{a} \in (\ell_\nu^1)^L$ . Following the steps of Section 8.1 suggests that we restrict our attention to a finite number of terms in the sequence space. Because of the complexity of working with vector spaces we introduce additional notation.

Let  $\pi^l: (\ell_\nu^1)^L \rightarrow \ell_\nu^1$  denote projection onto components, e.g.,

$$\pi^l(a) = a_l = \{a_{l,n}\}_{n=0}^\infty.$$

Define  $\pi_N: \ell_\nu^1 \rightarrow \ell_\nu^1$  by

$$(\pi_N a)_n = \begin{cases} a_n & \text{if } n \leq N \\ 0 & \text{otherwise} \end{cases}$$

and  $\pi_N^L: (\ell_\nu^1)^L \rightarrow (\ell_\nu^1)^L$  by  $\pi_N^L a = (\pi_N a_1, \dots, \pi_N a_L)$ . We define the projections onto the complement by

$$\pi_{N,\infty} := I - \pi_N \quad \text{and} \quad \pi_{N,\infty}^L := I - \pi_N^L.$$

Since we have set  $X = (\ell_\nu^1)^L$ , we denote the images of these projections by

$$\begin{aligned} X_N &:= \{a \in (\ell_\nu^1)^L \mid a_{l,n} = 0 \text{ if } n \geq N+1\} \\ X_{N,\infty} &:= \{a \in (\ell_\nu^1)^L \mid a_{l,n} = 0 \text{ if } 0 \leq n \leq N\}. \end{aligned}$$

We identify  $X_N$  with  $(\mathbb{R}^{(N+1)})^L$  via the bijection  $\iota_N^L: X_N \rightarrow (\mathbb{R}^{(N+1)})^L$  defined by

$$\iota_N^L a := \{a_{l,n} \mid 1 \leq l \leq L, 0 \leq n \leq N\}.$$

*Remark 8.2.1.* Technically the operators  $\pi^l$ ,  $\pi_N$ ,  $\pi_N^L$ ,  $\iota_N^L$  and the spaces  $X_N$  and  $X_{N,\infty}$  depend upon  $\nu$ . For simplicity of expression we have suppressed this dependence and instead will remark on it when necessary.

We return to the identification of an approximate solution  $\bar{a} \in (\ell_\nu^1)^L$ . Define  $F^{(N)}: (\mathbb{R}^{(N+1)})^L \rightarrow (\mathbb{R}^{(N+1)})^L$  by

$$F^{(N)}(a^{(N)}) = \iota_N^L \pi_N^L F(\iota_N^{-1} a^{(N)}) \quad \text{for } a^{(N)} \in (\mathbb{R}^{(N+1)})^L \quad (8.16)$$

where  $\pi_N^L: (\ell_\nu^1)^L \rightarrow (\ell_\nu^1)^L$ . Let  $\bar{a}^{(N)}$  be a numerical solution to the finite dimensional problem  $F^{(N)}(a) = 0$ . Set

$$\bar{a} := (\iota_N^L)^{-1} \bar{a}^{(N)} \in (\ell_\nu^1)^L. \quad (8.17)$$

The third step in the application of Theorem 7.6.2 is the choice of a linear operator  $A^\dagger: X \rightarrow Y$  that approximates  $DF(\bar{a})$ . Recall that  $\{e_m \in \ell_\nu^1 \mid m \in \mathbb{N}, \text{ and } e_{m,n} = \delta_{m,n}\}$  is a Schauder basis for  $\ell_\nu^1$ . For each  $l = 1, \dots, L$  we describe how the derivative acts on  $e_m$ :

$$(D_j F_l(a) e_m)_n = \begin{cases} \delta_{j,l} \delta_{m,0} & \text{for } n = 0, \\ -(D_j \phi_l(a))_{n-1-m} & \text{for } n \geq 1, m \leq n-1, \\ k \delta_{j,l} & \text{for } n = m. \end{cases} \quad (8.18)$$

Conceptually, it may be helpful to think of this as an  $L \times L$  matrix of infinite dimensional matrices, i.e.,

$$DF(a) = \begin{bmatrix} D_1 F_1(a) & \cdots & D_L F_1(a) \\ \vdots & \ddots & \vdots \\ D_1 F_L(a) & \cdots & D_L F_L(a) \end{bmatrix}$$

where a more detailed representation along the lines of (8.10) takes the form

$$\left[ \begin{array}{ccc|ccc|ccc} D_1 F_1^{(N)}(\bar{a}) & \mathbf{0} & \dots & & D_1 F_1^{(N)}(\bar{a}) & \mathbf{0} & \dots & & D_L F_1^{(N)}(\bar{a}) & \mathbf{0} & \dots \\ * & N+1 & \dots & \dots & * & 0 & \dots & \dots & * & 0 & \dots \\ \vdots & \vdots & \ddots & & \vdots & \vdots & \ddots & & \vdots & \vdots & \ddots \\ \hline & \vdots & & \ddots & & \vdots & & & & \vdots & \\ D_l F_l^{(N)}(\bar{a}) & \mathbf{0} & \dots & & D_l F_l^{(N)}(\bar{a}) & \mathbf{0} & \dots & & D_L F_l^{(N)}(\bar{a}) & \mathbf{0} & \dots \\ * & 0 & \dots & \dots & * & N+1 & \dots & \dots & * & 0 & \dots \\ \vdots & \vdots & \ddots & & \vdots & \vdots & \ddots & & \vdots & \vdots & \ddots \\ \hline & \vdots & & & & \vdots & & \ddots & & \vdots & \\ D_1 F_L^{(N)}(\bar{a}) & \mathbf{0} & \dots & & D_l F_L^{(N)}(\bar{a}) & \mathbf{0} & \dots & & D_L F_L^{(N)}(\bar{a}) & \mathbf{0} & \dots \\ * & 0 & \dots & \dots & * & 0 & \dots & \dots & * & N+1 & \dots \\ \vdots & \vdots & \ddots & & \vdots & \vdots & \ddots & & \vdots & \vdots & \ddots \end{array} \right]$$

To rigorously decompose  $DF$  into finite and infinite dimensional operators we use the identification of  $X_N$  with  $(\mathbb{R}^{(N+1)})^L$  via  $\iota_N^L$ , as well as the (finite dimensional) Jacobian  $DF^{(N)}$  evaluated at  $\bar{a}^{(N)}$ , to define the operator  $A_N^\dagger : X_N \rightarrow Y_N$  by

$$A_N^\dagger := \iota_N^{-1} \left( DF^{(N)} \left( \bar{a}^{(N)} \right) \right) \iota_N.$$

Since we have set  $Y = (\ell_{\nu'}^1)^L$ , for some  $\nu' \in (0, \nu)$ , we denote the range of  $\pi_N$  acting on  $Y$  by  $Y_N$ . As in the one-dimensional case of Section 8.1 to define the infinite part of  $A^\dagger$  we ignore the off diagonal terms, i.e., we define  $A_\infty^\dagger : X_{N,\infty} \rightarrow Y_{N,\infty}$  by

$$(\pi^l A_\infty^\dagger b)_n = n(\pi^l b)_n, \quad \text{for } l = 1, \dots, L. \quad (8.19)$$

Set

$$A^\dagger b := A_N^\dagger \pi_N^L b + A_\infty^\dagger \pi_{N,\infty}^L b \quad \text{for } b \in X. \quad (8.20)$$

Conceptually we have

$$A^\dagger = \left[ \begin{array}{cc|c|cc|c|c} D_1 F_1^{(N)}(\bar{a}) & \mathbf{0} & \dots & D_l F_1^{(N)}(\bar{a}) & \mathbf{0} & \dots & D_L F_1^{(N)}(\bar{a}) & \mathbf{0} \\ \mathbf{0} & \Lambda & & \mathbf{0} & \mathbf{0} & & \mathbf{0} & \mathbf{0} \\ \hline \vdots & & \ddots & \vdots & & & \vdots & \\ D_1 F_l^{(N)}(\bar{a}) & \mathbf{0} & \dots & D_l F_l^{(N)}(\bar{a}) & \mathbf{0} & \dots & D_L F_l^{(N)}(\bar{a}) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & & \mathbf{0} & \Lambda & & \mathbf{0} & \mathbf{0} \\ \hline \vdots & & & \vdots & & \ddots & \vdots & \\ D_1 F_L^{(N)}(\bar{a}) & \mathbf{0} & \dots & D_l F_L^{(N)}(\bar{a}) & \mathbf{0} & \dots & D_L F_L^{(N)}(\bar{a}) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & & \mathbf{0} & \mathbf{0} & & \mathbf{0} & \Lambda \end{array} \right]$$

where  $\Lambda$  is an infinite diagonal matrix with diagonal entries  $n$ , starting at  $N + 1$ .

The fourth step in the application of Theorem 7.6.2 is the choice of a linear operator  $A: Y \rightarrow X$  that approximates  $(A^\dagger)^{-1}$ . To gain intuition as to what this operator should be observe that reorganizing bases of  $X$  and  $Y$  allows us to rewrite  $A^\dagger$  as

$$\left[ \begin{array}{ccccccccc} D_1 F_1^{(N)}(\bar{a}) & \dots & D_l F_1^{(N)}(\bar{a}) & \dots & D_L F_1^{(N)}(\bar{a}) & \mathbf{0} & \dots & \mathbf{0} & \dots & \mathbf{0} \\ \vdots & & \vdots & & \vdots & & & & & \\ D_1 F_l^{(N)}(\bar{a}) & \dots & D_l F_l^{(N)}(\bar{a}) & \dots & D_L F_l^{(N)}(\bar{a}) & \mathbf{0} & \dots & \mathbf{0} & \dots & \mathbf{0} \\ \vdots & & \vdots & & \vdots & & & & & \\ D_1 F_L^{(N)}(\bar{a}) & \dots & D_l F_L^{(N)}(\bar{a}) & \dots & D_L F_L^{(N)}(\bar{a}) & \mathbf{0} & \dots & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \dots & \mathbf{0} & \dots & \mathbf{0} & \Lambda & \dots & \mathbf{0} & \dots & \mathbf{0} \\ \vdots & & \vdots & & \vdots & \vdots & \ddots & \vdots & & \vdots \\ \mathbf{0} & \dots & \mathbf{0} & \dots & \mathbf{0} & \mathbf{0} & \dots & \Lambda & \dots & \mathbf{0} \\ \vdots & & \vdots & & \vdots & \vdots & & \vdots & \ddots & \vdots \\ \mathbf{0} & \dots & \mathbf{0} & \dots & \mathbf{0} & \mathbf{0} & \dots & \mathbf{0} & \dots & \Lambda \end{array} \right]$$

where there are  $L$  copies of  $\Lambda$ . More compactly we have

$$\left[ \begin{array}{cccccc} DF^{(N)}(\bar{a}) & \mathbf{0} & \dots & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \Lambda & \dots & \mathbf{0} & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots & & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \Lambda & \dots & \mathbf{0} \\ \vdots & \vdots & & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{0} & \dots & \Lambda \end{array} \right]$$



which suggests that

$$A = \begin{bmatrix} A^{(N)} & \mathbf{0} & \cdots & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \Lambda^{-1} & \cdots & \mathbf{0} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots & & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \Lambda^{-1} & \cdots & \mathbf{0} \\ \vdots & \vdots & & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & \cdots & \Lambda^{-1} \end{bmatrix}$$

where  $A^{(N)}$  is a reasonable approximation to  $DF^{(N)}(\bar{a})^{-1}$ . Rewriting this in the original basis gives us a matrix of the form

$$A^\dagger = \left[ \begin{array}{cc|c|cc|c|c} A_{1,1}^{(N)}(\bar{a}) & \mathbf{0} & \cdots & A_{l,1}^{(N)}(\bar{a}) & \mathbf{0} & \cdots & A_{L,1}^{(N)}(\bar{a}) & \mathbf{0} \\ \mathbf{0} & \Lambda^{-1} & & \mathbf{0} & \mathbf{0} & & \mathbf{0} & \mathbf{0} \\ \hline \vdots & & \ddots & \vdots & & & \vdots & \\ \hline A_{1,l}^{(N)}(\bar{a}) & \mathbf{0} & \cdots & A_{l,l}^{(N)}(\bar{a}) & \mathbf{0} & \cdots & A_{L,l}^{(N)}(\bar{a}) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & & \mathbf{0} & \Lambda^{-1} & & \mathbf{0} & \mathbf{0} \\ \hline \vdots & & & \vdots & & \ddots & \vdots & \\ \hline A_{1,L}^{(N)}(\bar{a}) & \mathbf{0} & \cdots & A_{l,L}^{(N)}(\bar{a}) & \mathbf{0} & \cdots & A_{L,L}^{(N)}(\bar{a}) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & & \mathbf{0} & \mathbf{0} & & \mathbf{0} & \Lambda^{-1} \end{array} \right]$$

The matrix manipulations are meant to provide intuition concerning the choice of  $A^\dagger$  and  $A$ . To obtain the necessary formulas we recall that (7.26) provides an explicit expression for the matrix  $DF^{(N)}(\bar{a}^{(N)})$ . We use this to identify the desired approximate inverse  $A^{(N)}$ . For example, if  $N$  is not too large, then we can numerically compute an approximate inverse to  $DF^{(N)}(\bar{a}^{(N)})$ . Given  $A^{(N)}$  define  $A_N : Y_N \rightarrow X_N$  by

$$A_N = \iota_N^{-1} A^{(N)} \iota_N$$

and define  $A_\infty : Y_\infty \rightarrow X_\infty$  by

$$(\pi^l A_\infty b)_n = \frac{1}{k} (\pi^l b)_n, \quad \text{for } l = 1, \dots, L.$$

Finally, define

$$Ab = A_N \pi_N b + A_\infty \pi_{N,\infty} b \quad \text{for } b \in Y. \quad (8.21)$$

We leave it to the reader to check that since  $b \in \ell_{\nu'}^1$  and  $\nu' < \nu$ ,  $A : \ell_{\nu'}^1 \rightarrow \ell_\nu^1$ .

A direct application of Proposition 7.6.5 implies that if we identify a value  $r_0$  at which the radii polynomial  $p(r_0) < 0$ , then  $A$  is injective.

**Theorem 8.2.2.** Fix  $N$  and  $\nu$ . Let  $F^{(N)}: (\mathbb{R}^{N+1})^L \rightarrow (\mathbb{R}^{N+1})^L$  be defined by 8.16. Let  $\bar{a}^{(N)} = (\bar{a}_0, \dots, \bar{a}_N)$  be an approximate solution to  $F^{(N)}(a) = 0$  and let  $\bar{a}$  be given by (8.17). Let  $A^{(N)}$  be an approximate inverse of  $DF^{(N)}(\bar{a})$ .

Set

$$Y_0 \stackrel{\text{def}}{=} \max(Y_0^{(1)}, \dots, Y_0^{(n)})$$

where

$$Y_0^{(l)} \stackrel{\text{def}}{=} \sum_{n=0}^N \left| \sum_{j=1}^n \left( A_{l,j}^{(N)} F_j^{(N)}(\bar{a}^{(N)}) \right)_n \right| \nu^n + \nu \sum_{n=N}^{MN} \left| \frac{1}{n+1} \phi_l(\bar{a})_n \right| \nu^n, \quad \text{for } l = 1, \dots, L.$$

Set

$$Z_0 \stackrel{\text{def}}{=} \max_{1 \leq l \leq L} \sum_{j=1}^L Z_0^{(l,j)}. \quad (8.22)$$

where

$$Z_0^{(l,j)} \stackrel{\text{def}}{=} \max_{0 \leq n \leq N} \frac{1}{\nu^n} \sum_{0 \leq k \leq N} \left| \left( B_{l,j}^{(N)} \right)_{k,n} \right| \nu^k.$$

and

$$B_{l,j}^{(N)} \stackrel{\text{def}}{=} \delta_{l,j} I_{\mathbb{R}^{N+1}} - \sum_{m=1}^L A_{l,m}^{(N)} D_j F_m^{(N)}(\bar{a}^{(N)}) \in M_{N+1}(\mathbb{R}).$$

Set

$$Z_1 \stackrel{\text{def}}{=} \frac{\nu}{N+1} \max_{1 \leq l \leq L} \sum_{j=1}^n Z_1^{(l,j)}, \quad (8.23)$$

where

$$Z_1^{(l,j)} := \sum_{k=0}^{N(M-1)} |(D_j \phi_l(\bar{a}))_k| \nu^k,$$

for  $1 \leq l, j \leq L$ .

Fix  $r_* > 0$  and set

$$Z_2 := \nu \max_{1 \leq l \leq L} \sum_{j=1}^n Z_2^{(l,j)}, \quad (8.24)$$

where

$$Z_2^{(l,j)} := \sum_{m'=1}^n K_{l,m'} \sum_{m=1}^n \zeta_{m',m,j}(r_*) \quad \text{for } 1 \leq l, j \leq L,$$

$$K_{j,j'} := \max \left\{ \max_{1 \leq k' \leq N} \frac{1}{\nu^{k'}} \sum_{k=0}^N |(A_{j,j'}^{(N)})_{k,k'}| \nu^k, \frac{\delta_{j,j'}}{N+1} \right\}.$$

and

$$\zeta_{j,m,j'}(r) := \sum_{|\alpha|=2}^M |(c_\alpha)_j| \alpha_{j'} (\alpha_m - \delta_{j',m}) \prod_{l=1}^L (\|\bar{a}_l\|_{1,\nu} + r)^{\alpha_l - \delta_{j',l} - \delta_{m,l}}, \quad \text{for } 1 \leq j, j', m \leq L.$$

Then,  $Y_0$ ,  $Z_0$ ,  $Z_1$ , and  $Z_2$  satisfy the hypotheses of Theorem 7.6.2.

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Through (8.15) we have reduced the problem of proving existence of solutions of general IVPs to solving the operator equation

$$F(a) = (F_1(a), \dots, F_n(a)) = 0$$

in the Banach space  $X = (\ell_\nu^1)^n$ .

**Lemma 8.2.3.** *Let  $\nu > 0$ . When  $a \in (\ell_\nu^1)^n$  satisfies  $F(a) = 0$ , then  $x(t)$  defined by (8.14) solves the initial value problem (8.13) for  $t \in (-\nu, \nu)$ .*

*Proof.* For each  $j = 1, \dots, n$ , since  $a_j \in \ell_\nu^1$  the power series in (8.14) has radius of convergence at least  $\nu$ . Let  $x_j(t)$  be defined by (8.14) for  $t \in (-\nu, \nu)$ , then  $x_j(0) = (a_j)_0 = (x_0)_j$  since  $F_j(a)_0 = 0$ , hence  $x(t)$  satisfies the initial data. Furthermore, in the interior of the interval of convergence, differentiation of power series can be done term by term, hence  $x'_j(t) = \sum_{k=1}^{\infty} k(a_j)_k t^{k-1}$  for all  $t \in (-\nu, \nu)$ . From Lemma 8.4.7 and the relation (??) between the nonlinearities  $f$  and  $\phi$ , we infer that  $f_j(x(t)) = \sum_{k=0}^{\infty} (\phi_j(a))_k t^k$  for all  $t \in (-\nu, \nu)$ . It now follows from  $(F_j(a))_k = 0$  for all  $j = 1, \dots, n$  and  $k \geq 1$  that  $x'(t) = f(x(t))$  for  $t \in (-\nu, \nu)$ , hence  $x(t)$  solves the initial value problem on this time interval.  $\square$

In the analysis of the estimates we will encounter the component-wise *shift* operator  $S : \ell_\nu^1 \rightarrow \ell_\nu^1$  defined by

$$(Sa)_k = \begin{cases} 0 & \text{if } k = 0 \\ a_{k-1} & \text{if } k \geq 1. \end{cases} \quad (8.25)$$

It turns out that it is also convenient to introduce another component-wise operator  $\Lambda_N : X_\infty^{1,\nu} \rightarrow X_\infty^{1,\nu}$  defined by

$$(\Lambda_N a)_k = \begin{cases} 0 & \text{if } 0 \leq k \leq N, \\ \frac{1}{k} a_k & \text{if } k \geq N+1. \end{cases} \quad (8.26)$$

The proof of the following lemma concerning the norms of these operators is left to the reader.

**Lemma 8.2.4.** *For fixed  $N$ ,*

$$\|\pi_N\|_{B(X)} = \|\pi_\infty\|_{B(X)} = 1, \quad \|S\|_{B(\ell_\nu^1)} = \nu, \quad \|\Lambda_N\|_{B(X_\infty^{1,\nu})} = \frac{1}{N+1}.$$

Finally, we note that since  $X_\infty^{1,\nu}$  is a subspace of  $\ell_\nu^1$  the operator  $\Lambda_N$  may also be interpreted as mapping  $X_\infty^{1,\nu}$  into  $\ell_\nu^1$ . Likewise,  $\pi_\infty^{1,\nu}$  may be interpreted as mapping from  $\ell_\nu^1$  to  $X_\infty^{1,\nu}$ . Such trivial embedding properties will not be commented on in the remainder of this chapter.

### 8.2.1 Fixed point formulation

In this section, as in Section 8.1, we solve the problem  $F(a) = 0$  by assuming that we have an approximate solution  $\bar{a} \in X_N$  and introducing an associated fixed point operator of the form  $T(a) = a - AF(a)$ . Since we are in the infinite dimensional setting we do this by finding an operator  $A^\dagger$  that approximates  $DF(\bar{a})$  and then choosing  $A$  to be an approximate inverse of  $A^\dagger$ .

Defining  $A^\dagger$  requires understanding the structure of the Fréchet derivative  $DF(\bar{a})$ , which involves, recalling (8.15), computing  $D\phi_i(a)$ , for each  $i = 1, \dots, n$ . It follows from (??) and the chain rule that

$$D\phi_j(a)b = \mathcal{T} \sum_{j'=1}^n \partial_{j'} f_j(\mathcal{T}^{-1}a_1, \dots, \mathcal{T}^{-1}a_n) \mathcal{T}^{-1}b_{j'},$$

where  $b = (b_1, \dots, b_n)$  with  $b_j \in \ell_\nu^1$ . Hence, using the same notational convention for convolution products as in (7.25), we can represent  $D_{a_{j'}}\phi_j(a)$  by the convolution product

$$G_{j,j'}(a) := \sum_{|\alpha|=1}^M (c_\alpha)_j \alpha_{j'} a^{\alpha - \epsilon_{j'}},$$

so that

$$D\phi_j(a)b = \sum_{j'=1}^n G_{j,j'}(a) * b_{j'}.$$

For simplicity of notation, we will throughout identify  $D_{a_{j'}}\phi_j(a)$  with  $G_{j,j'}(a)$  and write

$$D_{a_{j'}}\phi_j(a) = \sum_{|\alpha|=1}^M (c_\alpha)_j \alpha_{j'} a^{\alpha - \epsilon_{j'}},$$

as well as

$$D\phi_j(a)b = \sum_{j'=1}^n D_{a_{j'}}\phi_j(a)b_{j'} = \sum_{j'=1}^n \sum_{|\alpha|=1}^M (c_\alpha)_j \alpha_{j'} a^{\alpha - \epsilon_{j'}} * b_{j'}. \quad (8.27)$$

For sake of simplicity of the presentation, we denote by  $D_{j'} = D_{a_{j'}}$  the Fréchet differentiation with respect to the variable  $a_{j'}$ . The derivative  $D_{j'}F_j(a)$  can be characterized by its action on natural basis vectors  $\{e_m\}_{m=0}^\infty$  of  $\ell_\nu^1$ , defined by  $(e_m)_k = \delta_{m,k}$ :

$$(D_{j'}F_j(a)e_m)_k = \begin{cases} \delta_{j',j}\delta_{m,0} & \text{for } k = 0, \\ -(D_{j'}\phi_j(a))_{k-1-m} & \text{for } k \geq 1, m \leq k-1, \\ k\delta_{j',j} & \text{for } k = m. \end{cases} \quad (8.28)$$

Here we have used the derivative notation  $D_{j'}\phi_j(a)$  from (8.27).

Using the identification of  $X_N$  and  $\mathbb{R}^{n(N+1)}$  via  $\iota_N$ , as well as the (finite dimensional) Jacobian  $DF^{(N)}$ , we define the operator  $A_N^\dagger : X_N \rightarrow X_N$  by

$$A_N^\dagger := \iota_N^{-1}DF^{(N)}(\bar{a}^{(N)})\iota_N,$$

and it follows that

$$A_N^\dagger = \pi_N DF(\bar{a}) \quad \text{for any } \bar{a} \in X_N. \quad (8.29)$$

We also define the operator  $A_\infty^\dagger : X_\infty \rightarrow X_\infty$  by

$$(\pi^j A_\infty^\dagger b)_k = k(\pi^j b)_k, \quad \text{for } j = 1, \dots, n. \quad (8.30)$$

We now set

$$A^\dagger b = A_N^\dagger \pi_N b + A_\infty^\dagger \pi_\infty b. \quad (8.31)$$

Next we turn our attention to defining an approximate inverse  $A$  of  $A^\dagger$ . We note that the matrix  $DF^{(N)}(\bar{a}^{(N)})$  can be determined using the explicit expression (8.28). With  $A^{(N)}$  a numerically computed approximate inverse of the matrix  $DF^{(N)}(\bar{a}^{(N)})$ , we set  $A_N : X_N \rightarrow X_N$  to be the approximate inverse of  $A_N^\dagger$  defined by

$$A_N = \iota_N^{-1}A^{(N)}\iota_N.$$

We define  $A_\infty : X_\infty \rightarrow X_\infty$  by

$$(\pi^j A_\infty b)_k = \frac{1}{k}(\pi^j b)_k, \quad \text{for } j = 1, \dots, n.$$

so that  $A_\infty = (A_\infty^\dagger)^{-1}$ , hence it makes sense to write  $A_\infty^{-1}$  instead of  $A_\infty^\dagger$ . We then define

$$Ab = A_N \pi_N b + A_\infty \pi_\infty b. \quad (8.32)$$

It is essential to our fixed point argument that  $A$  is injective, as that guarantees that a fixed point of  $T(x) = x - AF(x)$  correspond to a zero of  $F$ . The linear map  $A$  is defined on the set of all sequences  $(\mathbb{R}^\mathbb{N})^n$ . The map  $A$  block-decomposes and its action can be represented by a matrix  $A^{(N)}$  on  $\pi_N(\mathbb{R}^\mathbb{N})^n$  whereas it acts diagonally on the tail  $\pi_\infty(\mathbb{R}^\mathbb{N})^n$ . Since this diagonal action is bijective, the map  $A$  is injective if and only if  $A^{(N)}$  is an invertible matrix. The following lemma gives a criterium for injectivity of linear maps  $A$  of this form. It is formulated in quite general terms, so that we can reuse it later.

**Lemma 8.2.5.** *Let  $A$  be as above. Let  $X$  be a linear subspace of  $(\mathbb{R}^{\mathbb{N}})^n$  equipped with a norm  $\|\cdot\|_X$ . Let  $B : X \rightarrow (\mathbb{R}^{\mathbb{N}})^n$  be a linear map such that the composition  $AB$  maps  $X$  to itself and  $\|I - AB\|_{B(X)} < 1$ . Then  $A$  is injective.*

*Proof.* As argued above, it suffices to show that  $A^{(N)}$  is an invertible matrix in  $M_{N_*}(\mathbb{R})$ , where  $N_* = n(N + 1)$ . The proof of invertibility of  $A^{(N)}$  is heavy on notation, but in fact quite straightforward. We will identify a natural approximate inverse matrix  $B^{(N)}$  and an induced norm on  $\mathbb{R}^{N_*}$  and then show that  $\|I_{N_*} - A^{(N)}B^{(N)}\|_{B(\mathbb{R}^{N_*})} < 1$  with  $I_{N_*}$  the identity on  $\mathbb{R}^{N_*}$ , implying that  $A^{(N)}$  is invertible, see Exercise 2.7.1.

We define the matrix  $B^{(N)} = \iota_N \pi_N B \iota_N^{-1} \in M_{N_*}(\mathbb{R})$ . We introduce a norm on  $\mathbb{R}^{N_*}$  defined by  $\|x^{(N)}\|_{\mathbb{R}^{N_*}} = \|\iota_N^{-1} x^{(N)}\|_X$ , so that  $\iota_N$  is an isometry between  $X_N$  and  $\mathbb{R}^{N_*}$ . We assert that  $\|I_{N_*} - A^{(N)}B^{(N)}\|_{B(\mathbb{R}^{N_*})} < 1$ . Namely, we first rewrite

$$\begin{aligned} I_{N_*} - A^{(N)}B^{(N)} &= I_{N_*} - \iota_N \iota_N^{-1} A^{(N)} \iota_N \pi_N B \iota_N^{-1} \\ &= I_{N_*} - \iota_N A_N \pi_N B \iota_N^{-1} \\ &= I_{N_*} - \iota_N \pi_N A B \iota_N^{-1} \\ &= \iota_N \pi_N I_X \iota_N^{-1} - \iota_N \pi_N A B \iota_N^{-1} \\ &= \iota_N \pi_N (I_X - AB) \iota_N^{-1}. \end{aligned}$$

Writing  $I_X = I$ , we then infer that

$$\begin{aligned} \|I_{N_*} - A^{(N)}B^{(N)}\|_{B(\mathbb{R}^{N_*})} &= \|\iota_N \pi_N (I - AB) \iota_N^{-1}\|_{B(\mathbb{R}^{N_*})} \\ &= \|\pi_N (I - AB)\|_{B(X_N)} \\ &\leq \|\pi_N (I - AB)\|_{B(X, X_N)} \\ &\leq \|\pi_N\|_{B(X, X_N)} \|I - AB\|_{B(X)} \\ &= \|I - AB\|_{B(X)} < 1. \end{aligned} \quad \square$$

### Decomposition into components

Although strictly speaking the ordering of the variables in  $\mathbb{R}^{n(N+1)}$  is arbitrary, it is useful to keep in mind the identification  $(\mathbb{R}^{N+1})^n \simeq \mathbb{R}^{n(N+1)}$ :

$$a^{(N)} = (a_1^{(N)}, \dots, a_n^{(N)}) \quad \text{and} \quad F^{(N)}(a^{(N)}) = (F_1^{(N)}(a^{(N)}), \dots, F_n^{(N)}(a^{(N)})),$$

with  $a_j^{(N)} \in \mathbb{R}^{N+1}$  and  $F_j^{(N)} \in \mathbb{R}^{N+1}$ .

We now look more concretely at how  $A^\dagger$  and  $A$  act *component-wise*. This is essential for deriving some of the estimates, and also convenient for implementation purposes. We write  $b = (b_1, \dots, b_n)$  and  $b^{(N)} = \iota_N \pi_N b = (b_1^{(N)}, \dots, b_n^{(N)})$ . We decompose  $A^\dagger$  as

$$\pi^j A^\dagger b = \sum_{j'=1}^n A_{j,j'}^\dagger b_{j'},$$

with

$$(A_{j,j'}^\dagger b_{j'})_k = \begin{cases} \left( D_{j'} F_j^{(N)}(\bar{a}^{(N)}) b_{j'}^{(N)} \right)_k, & 0 \leq k \leq N \\ \delta_{j,j'} \frac{1}{k} (b_{j'})_k, & k > N. \end{cases}$$

Here  $D_{j'} F_j^{(N)}$  is shorthand for  $D_{a_{j'}}^{(N)} F_j^{(N)}$ . One may view  $A_{j,j'}^\dagger$  as an infinite matrix

$$A_{j,j'}^\dagger \stackrel{\text{def}}{=} \begin{bmatrix} D_{j'} F_j^{(N)}(\bar{a}^{(N)}) & \mathbf{0}^T & \mathbf{0}^T & \mathbf{0}^T \\ \mathbf{0} & (N+1)\delta_{j,j'} & 0 & 0 \\ \mathbf{0} & 0 & (N+2)\delta_{j,j'} & 0 \\ \mathbf{0} & 0 & 0 & \ddots \end{bmatrix},$$

where  $\mathbf{0} = (0, 0, \dots, 0, 0) \in \mathbb{R}^{N+1}$ . The infinite diagonal is thus present only when  $i = j$ .

Next, we note that  $DF^{(N)}(\bar{a}^{(N)})$  can be written as a decomposition in block matrix

$$DF^{(N)}(\bar{a}^{(N)}) = \begin{bmatrix} D_1 F_1^{(N)}(\bar{a}^{(N)}) & \cdots & D_n^{(N)} F_1(\bar{a}^{(N)}) \\ \vdots & \ddots & \vdots \\ D_1 F_1^{(N)}(\bar{a}^{(N)}) & \cdots & D_n^{(N)} F_n(\bar{a}^{(N)}) \end{bmatrix}.$$

Similarly,  $A^{(N)}$  can be written as

$$A^{(N)} \stackrel{\text{def}}{=} \begin{bmatrix} A_{1,1}^{(N)} & \cdots & A_{1,n}^{(N)} \\ \vdots & \ddots & \vdots \\ A_{n,1}^{(N)} & \cdots & A_{n,n}^{(N)} \end{bmatrix} \in M_{n(N+1)}(\mathbb{R}). \quad (8.33)$$

We emphasize that  $A^{(N)}$  is an approximate inverse of  $DF^{(N)}(\bar{a}^{(N)})$ , but that  $A_{j,j'}^{(N)}$  is in general *not* an approximate inverse of  $D_{a_{j'}} F_j^{(N)}(\bar{a}^{(N)})$ . Indeed,  $A_{j,j'}^{(N)}$  is just a block in the matrix obtained by numerically inverting (a numerical approximation of)  $DF^{(N)}(\bar{a}^{(N)})$ .

In term of components we can write

$$\pi^j A b = \sum_{j'=1}^n A_{j,j'} b_{j'},$$

where

$$(A_{j,j'} b_{j'})_k = \begin{cases} \left( A_{j,j'}^{(N)} b_{j'}^{(N)} \right)_k & 0 \leq k \leq N, \\ \delta_{j,j'} \frac{1}{k} (b_{j'})_k & k > N. \end{cases}$$

In particular, the operator  $A_\infty$  can be written as

$$\pi^j A_\infty b = \Lambda_N \pi_\infty b_j,$$

with  $\Lambda_N$  defined in (8.26). A graphically intuitive representation of  $A$  is given by

$$A = \left[ \begin{array}{cc|cc|cc|cc} A_{1,1}^{(N)} & 0 & A_{1,2}^{(N)} & 0 & \cdots & \cdots & A_{1,n}^{(N)} & 0 \\ 0 & \Lambda_N & 0 & 0 & \cdots & \cdots & 0 & 0 \\ \hline A_{2,1}^{(N)} & 0 & A_{2,2}^{(N)} & 0 & & & A_{2,n}^{(N)} & 0 \\ 0 & 0 & 0 & \Lambda_N & & & 0 & 0 \\ \hline \vdots & \vdots & & & \ddots & & \vdots & \vdots \\ \vdots & \vdots & & & & \ddots & \vdots & \vdots \\ \hline A_{n,1}^{(N)} & 0 & A_{n,2}^{(N)} & 0 & \cdots & \cdots & A_{n,n}^{(N)} & 0 \\ 0 & 0 & 0 & 0 & \cdots & \cdots & 0 & \Lambda_N \end{array} \right].$$

### Estimates

With  $A$  and  $A^\dagger$  defined in (8.32) and (8.31), each written in terms of components in Section 8.2.1, we are ready to define the Newton-like fixed point operator

$$T(a) = a - AF(a).$$

To apply Theorem 7.6.2, the first thing to check is that  $AF(a) \in X$  for all  $a \in X$ . The following results demonstrates that this is the case.

**Proposition 8.2.6.** *Let  $a \in X = (\ell_\nu^1)^n$ . Let  $F$  defined in (8.15). Let  $A$  be defined in (8.32). Then  $AF(a) \in X$ .*

*Proof.* Let  $a = (a_1, \dots, a_n) \in X$  be arbitrary, i.e.,  $\|a_j\|_{1,\nu} < \infty$  for  $1 \leq j \leq n$ . For each  $j = 1, \dots, n$  we estimate

$$\begin{aligned} \|\pi^j AF(a)\|_{1,\nu} &= \left\| \sum_{j'=1}^n A_{j,j'} F_{j'}(a) \right\|_{1,\nu} \\ &= \sum_{k=0}^N \left| \sum_{j'=1}^n \left( A_{j,j'}^{(N)} F_{j'}^{(N)}(\iota_N \pi_N a) \right)_k \right| \nu^k + \sum_{k>N} \frac{1}{k} |k(a_j)_k - (\phi_j(a))_{k-1}| \nu^k \\ &\leq \sum_{k=0}^N \left| \sum_{j'=1}^n \left( A_{j,j'}^{(N)} F_{j'}^{(N)}(\iota_N \pi_N a) \right)_k \right| \nu^k + \|a_j\|_{1,\nu} + \frac{\nu}{N+1} \|\phi_j(a)\|_{1,\nu} < \infty. \end{aligned}$$

since  $\phi(a) \in (\ell_\nu^1)^n$  for all  $a \in (\ell_\nu^1)^n$ . Hence,  $\|AF(a)\|_X = \max_{1 \leq j \leq n} \|\pi^j AF(a)\|_{1,\nu} < \infty$ .  $\square$

We now compute the bounds  $Y_0$ ,  $Z_0$ ,  $Z_1$  and  $Z_2$  satisfying (7.30), (7.31), (7.32) and (7.33), respectively.



**The bound  $Y_0$** 

**Lemma 8.2.7.** *Set*

$$Y_0^{(j)} \stackrel{\text{def}}{=} \sum_{k=0}^N \left| \sum_{j'=1}^n \left( A_{j,j'}^{(N)} F_{j'}^{(N)}(\bar{a}^{(N)}) \right)_k \right| \nu^k + \nu \sum_{k=N}^{MN} \left| \frac{1}{k+1} (\phi_j(\bar{a}))_k \right| \nu^k,$$

for  $j = 1, \dots, n$ , and

$$Y_0 \stackrel{\text{def}}{=} \max \left( Y_0^{(1)}, \dots, Y_0^{(n)} \right). \quad (8.34)$$

Then  $\|T(\bar{a}) - \bar{a}\| \leq Y_0$ .

*Proof.* We note that, for  $j = 1, \dots, n$ ,

$$(\pi^j T(\bar{a}) - \bar{a}_j)_k = \begin{cases} - \sum_{j'=1}^n \left( A_{j,j'}^{(N)} F_{j'}^{(N)}(\bar{a}^{(N)}) \right)_k & \text{for } 0 \leq k \leq N, \\ -\frac{1}{k} (\phi_j(\bar{a}))_{k-1} & \text{for } k > N. \end{cases}$$

To obtain a bound on  $(\pi^j T(\bar{a}) - \bar{a}_j)_k$  for large  $k$ , we note that

$$(\phi_j(\bar{a}))_k = 0 \quad \text{for all } k > MN, \quad (8.35)$$

since  $\bar{a} \in X_N$  and  $\phi$  (or rather  $f$ ) is a polynomial of degree  $M$ . It follows from (8.35) that we can estimate  $\|\pi^j [T(\bar{a}) - \bar{a}]\|_{\ell_\nu^1}$  by  $Y_0^{(j)}$ , and hence  $\|T(\bar{a}) - \bar{a}\|_X$  by  $Y_0$ .  $\square$

**The bound  $Z_0$** 

Let

$$\begin{aligned} B &\stackrel{\text{def}}{=} I - AA^\dagger \\ &= I - (A_N \pi_N + A_\infty \pi_\infty)(A_N^\dagger \pi_N + A_\infty^\dagger \pi_\infty) \\ &= (\pi_N + \pi_\infty)I - (A_N A_N^\dagger \pi_N + A_\infty A_\infty^\dagger \pi_\infty) \\ &= (I_{X_N} - A_N A_N^\dagger) \pi_N, \end{aligned}$$

since  $A_\infty^\dagger = A_\infty^{-1}$ . Here  $I_{X_N}$  is the identity on  $X_N$ . Using the decomposition notation, we set

$$B_{j,j'}^{(N)} \stackrel{\text{def}}{=} \delta_{j,j'} I_{\mathbb{R}^{N+1}} - \sum_{m=1}^n A_{j,m}^{(N)} D_{j'} F_m^{(N)}(\bar{a}^{(N)}) \in M_{N+1}(\mathbb{R}).$$

These matrices contain all nontrivial behaviour of the infinite dimensional linear operator  $B$ :

$$\pi^j B = \sum_{j'=1}^n \iota_N^{-1} B_{j,j'}^{(N)} \iota_N \pi_N \pi^{j'}.$$

**Lemma 8.2.8.** *Set*

$$Z_0^{(j,j')} \stackrel{\text{def}}{=} \max_{0 \leq k' \leq N} \frac{1}{\nu^{k'}} \sum_{0 \leq k \leq N} \left| (B_{j,j'}^{(N)})_{k,k'} \right| \nu^k,$$

and

$$Z_0 \stackrel{\text{def}}{=} \max_{1 \leq j \leq n} \sum_{j'=1}^n Z_0^{(j,j')}. \quad (8.36)$$

Then  $\|I - AA^\dagger\|_{B(X)} \leq Z_0$ .

*Proof.* For any  $h \in X$  with  $\|h\|_X \leq 1$  we obtain, using the notation  $h_{j'}^{(N)} = \iota_N \pi_N \pi^{j'} h$ ,

$$\|\pi^j(Bh)\|_{1,\nu} = \left\| \iota_N^{-1} \sum_{j'=1}^n B_{j,j'}^{(N)} h_{j'}^{(N)} \right\|_{1,\nu} \leq \sum_{j'=1}^n \|B_{j,j'}^{(N)}\|_{\nu,N} \quad \text{for } j = 1, \dots, n,$$

with the matrix norm from (8.12). We thus obtain  $\|\pi^j(Bh)\|_{1,\nu} \leq \sum_{j'=1}^n Z_0^{(j,j')}$  and

$$\|(I - AA^\dagger)h\|_X = \max_{1 \leq j \leq n} \|\pi^j(Bh)\|_{1,\nu} \leq Z_0.$$

Since  $h \in X$  with  $\|h\|_X \leq 1$  is arbitrary, this finishes the proof.  $\square$

### The bound $Z_1$

Recall that we look for a bound  $\|A[DF(\bar{a}) - A^\dagger]\|_{B(X)} \leq Z_1$ . Given arbitrary  $h \in X$  with  $\|h\|_X \leq 1$ , set

$$z \stackrel{\text{def}}{=} [DF(\bar{a}) - A^\dagger]h. \quad (8.37)$$

It follows from (8.15) that  $DF(a)$  is lower triangular. To be more precise, since  $\pi_N F(a) = \pi_N F(\pi_N a)$ , we have that  $\pi_N DF(a) \pi_\infty = 0$ . Furthermore, we recall from (8.29) that  $\pi_N DF(\bar{a}) \pi_N = A_N^\dagger \pi_N$ . This implies that

$$\begin{aligned} DF(\bar{a}) - A^\dagger &= (\pi_N + \pi_\infty) DF(\bar{a}) - (A_N^\dagger \pi_N + A_\infty^\dagger \pi_\infty) \\ &= \pi_N DF(\bar{a}) (\pi_N + \pi_\infty) - A_N^\dagger \pi_N + \pi_\infty DF(\bar{a}) - A_\infty^\dagger \pi_\infty \\ &= \pi_\infty DF(\bar{a}) - A_\infty^\dagger \pi_\infty. \end{aligned}$$

In particular,  $\pi_N z = 0$  and  $z = \pi_\infty z$ .

**Lemma 8.2.9.** *Set*

$$Z_1^{(j,j')} := \|D_{j'} \phi_j(\bar{a})\|_{1,\nu} = \sum_{k=0}^{N(M-1)} |(D_{j'} \phi_j(\bar{a}))_k| \nu^k,$$

for  $1 \leq j, j' \leq n$ , and

$$Z_1 \stackrel{\text{def}}{=} \frac{\nu}{N+1} \max_{1 \leq j \leq n} \sum_{j'=1}^n Z_1^{(j,j')}, \quad (8.38)$$

then  $\|A[DF(\bar{a}) - A^\dagger]\|_{B(X)} \leq Z_1$ .

*Proof.* For arbitrary  $h \in X$  with  $\|h\|_X \leq 1$  we define  $z \stackrel{\text{def}}{=} [DF(\bar{a}) - A^\dagger]h$ . We already argued that  $z = \pi_\infty z$ . From the definitions (8.28) and (8.30) of  $DF$  and  $A_\infty^\dagger$ , we observe that

$$(\pi^j z)_k = \begin{cases} 0 & k = 0, \dots, N, \\ -(D\phi_j(\bar{a})h)_{k-1} & k > N. \end{cases}$$

We then estimate

$$\begin{aligned} \|\pi^j(Az)\|_{1,\nu} &= \|A_{j,j}\pi_\infty\pi^j z\|_{1,\nu} = \sum_{k \geq N+1} \frac{1}{k} |(\pi^j z)_k| \nu^k \\ &\leq \frac{\nu}{N+1} \sum_{k \geq N+1} |(D\phi_j(\bar{a})h)_{k-1}| \nu^{k-1} \\ &\leq \frac{\nu}{N+1} \left\| \sum_{j'=1}^n D_{j'} \phi_j(\bar{a}) \pi^{j'} h \right\|_{1,\nu} \\ &\leq \frac{\nu}{N+1} \sum_{j'=1}^n \|D_{j'} \phi_j(\bar{a})\|_{B(\ell_\nu^1, \ell_\nu^1)} \|h\|_X. \end{aligned} \quad \square$$

### The bound $Z_2$

Recall that we look for the bound  $\|A[DF(b) - DF(\bar{a})]\|_{B(X)} \leq Z_2 r$ , for all  $b \in \overline{B_r(\bar{a})}$ . Given arbitrary  $h \in X$  with  $\|h\|_X \leq 1$  and arbitrary  $b \in \overline{B_r(\bar{a})}$ , set

$$\beta_{j,j'} \stackrel{\text{def}}{=} [D_{j'} F_j(b) - D_{j'} F_j(\bar{a})] \pi_{j'} h \quad \text{for } j, j' = 1, \dots, n. \quad (8.39)$$

By linearity of the first component of (8.15), we get that  $(\beta_{j,j'})_0$  for all  $j, j' = 1, \dots, n$ . We infer that

$$\beta_{j,j'} = -S[D_{j'} \phi_j(b) - D_{j'} \phi_j(\bar{a})] \pi_{j'} h, \quad (8.40)$$

with the shift operator  $S$  defined in (8.25).

Any element  $b$  of the ball  $\overline{B_r(\bar{a})}$  may be written as  $b = \bar{a} + r\tilde{h}$  with  $\tilde{h} \in \overline{B_1(0)}$ . Applying the Mean Value Theorem (Theorem 7.2.4) to (8.40) yields

$$\|\beta_{j,j'}\|_{1,\nu} \leq \sup_{\substack{a \in \overline{B_r(\bar{a})} \\ \tilde{h} \in \overline{B_1(0)}}} \left\| D(D_{j'} \phi_j(a) \pi^{j'} h) \tilde{h} \right\|_{1,\nu} \nu r, \quad (8.41)$$

where we have used that  $\|S\|_{B(\ell_\nu^1, \ell_\nu^1)} = \nu$ . Using the product notation for the convolution as before (cf. (8.27)), we write the second derivative as

$$D_{m,j'}^2 \phi_j(a) = \sum_{|\alpha|=2}^M (c_\alpha)_j \alpha_{j'} (\alpha_m - \delta_{j',m}) a^{\alpha - \epsilon_{j'} - \epsilon_m} \quad \text{for } m, j' = 1, \dots, n.$$

**Lemma 8.2.10.** *Let  $r_* > 0$ . For  $1 \leq j, j', m \leq n$ , let*

$$\zeta_{j,m,j'}(r) := \sum_{|\alpha|=2}^M |(c_\alpha)_j| \alpha_{j'} (\alpha_m - \delta_{j',m}) \prod_{l=1}^n (\|\bar{a}_l\|_{1,\nu} + r)^{\alpha_l - \delta_{j',l} - \delta_{m,l}},$$

and

$$K_{j,j'} := \max \left\{ \max_{1 \leq k' \leq N} \frac{1}{\nu^{k'}} \sum_{k=0}^N |(A_{j,j'}^{(N)})_{k,k'}| \nu^k, \frac{\delta_{j,j'}}{N+1} \right\}.$$

Set

$$Z_2^{(j,j')} := \sum_{m'=1}^n K_{j,m'} \sum_{m=1}^n \zeta_{m',m,j'}(r_*)$$

for  $1 \leq j, j' \leq n$ , and

$$Z_2 := \nu \max_{1 \leq j \leq n} \sum_{j'=1}^n Z_2^{(j,j')}, \quad (8.42)$$

then  $\|A[DF(b) - DF(\bar{a})]\|_{B(X)} \leq Z_2 r$ , for all  $b \in \overline{B_r(\bar{a})}$  any  $r \in [0, r_*]$ .

*Proof.* Now, for any  $a \in \overline{B_r(\bar{a})}$  and  $\tilde{h}, h \in \overline{B_1(0)}$  the norm of the second derivative  $D_{m,j'}^2 \phi_j(a)[\pi^{j'} h, \pi^m \tilde{h}]$  may be estimated by  $\zeta_{j,m,j'}(r)$  so that

$$\left\| D(D_{j'} \phi_j(a) \pi^{j'} h) \tilde{h} \right\|_{1,\nu} \leq \sum_{m=1}^n \zeta_{j,m,j'}(r) \quad \text{for any } a \in \overline{B_r(\bar{a})} \text{ and } \tilde{h}, h \in \overline{B_1(0)}. \quad (8.43)$$

We can now use the triangle inequality to estimate  $\|A[DF(b) - DF(\bar{a})]\|_{B(X)}$ . In terms of components we have

$$\pi^j A[DF(b) - DF(\bar{a})]h = \sum_{j'=1}^n \sum_{m=1}^n A_{j,m} \beta_{m,j'},$$

with  $\beta$  defined in (8.39). Using the bound (10.34) on  $\beta$ , combined with (8.43) and the fact that the operator norms  $\|A_{j,j'}\|_{B(\ell_\nu^1)}$  are given by  $K_{j,j'}$ , we see that

$$\left\| \sum_{m=1}^n A_{j,m} \beta_{m,j'} \right\|_{\ell_\nu^1} \leq \nu Z_2^{(j,j')} r,$$

for any  $r \leq r_*$ , where we have used that  $\zeta$  is increasing in  $r$ . Finally, we obtain

$$\|A[DF(b) - DF(\bar{a})]\|_{B(X)} \leq \max_{1 \leq j \leq N} \|\pi^j A[DF(b) - DF(\bar{a})]\|_{B(X, \ell_\nu^1)} \leq Z_2 r. \quad \square$$

### Conclusion

We have thus proven the following result.

**Theorem 8.2.11.** *Let  $N \in \mathbb{N}$  and  $\nu > 0$ . Let  $f$  be a polynomial given by (7.23). Let  $\bar{a} \in X_N$ . Let  $Y_0, Z_0, Z_1$  and  $Z_2(r_*)$  be defined by (8.34), (8.36), (8.38) and (8.42), respectively. Assume that  $r_0 \in (0, r_*)$  satisfies the inequality*

$$Z_2(r_*)r_0^2 + (Z_0 + Z_1 - 1)r_0 + Y_0 < 0. \quad (8.44)$$

*Then there is an  $\tilde{a} \in X$  with  $\|\tilde{a} - \bar{a}\|_X \leq r_0$  such that*

$$\tilde{x}(t) \stackrel{\text{def}}{=} \sum_{k=0}^{\infty} \tilde{a}_k t^k \quad (8.45)$$

*solves the initial value problem (8.13) for  $t \in (-\nu, \nu)$ .*

We note that injectivity of  $A$  follows from Lemma 8.2.5, since inequality (8.44) can only hold for  $r_0 > 0$  if  $Z_0 < 1$ , which implies that  $\|I - AA^\dagger\| \leq Z_0 < 1$ .

Concerning the error bound between the numerical approximation  $\bar{x}(t) = \sum_{k=0}^N \bar{a}_k t^k$  and the solution  $\tilde{x}(t)$  given in (8.45), we have

$$\sup_{t \in (-\nu, \nu)} \|\tilde{x}(t) - \bar{x}(t)\| \leq \max_{1 \leq j \leq n} \sum_{k=0}^{\infty} |(\tilde{a}_j - \bar{a}_j)_k| \nu^k = \|\tilde{a} - \bar{a}\|_X \leq r_0.$$

## 8.3 Initial value problems for the Lorenz system

Consider the vector field  $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$  given by the Lorenz system

$$f(x) = \begin{pmatrix} \sigma(x_2 - x_1) \\ \rho x_1 - x_2 - x_1 x_3 \\ -\beta x_3 + x_1 x_2 \end{pmatrix},$$

with classical parameter values  $\sigma = 10$ ,  $\rho = 28$  and  $\beta = 8/3$ . We applied the general method from Section 8.2 to the Lorenz system to obtain validated orbit segments for various initial data, see Figure 8.2. The time interval for all depicted segments is  $[-0.15, 0.15]$ . We set  $L = 0.15$  and chose computational parameters  $N = 200$ ,  $\nu = 1$  and  $r_* = 0.1$ .

## 8.4 Taylor series and the Taylor transform

From a computational point of view the representation of an analytic function by its Taylor expansion is very useful as

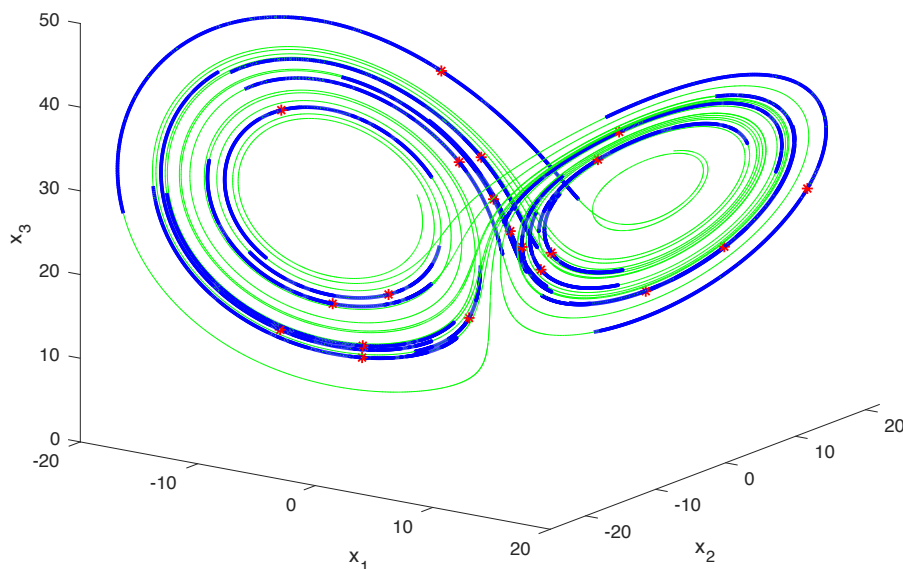


Figure 8.2: Twenty validated orbit segments near the Lorenz attractor: the figure illustrates a number of points near the Lorenz attractor (green points obtained by integrating a single initial condition for 25 time units). The blue curves represent one step of Taylor integration computed to order  $N = 200$  for different initial conditions near the attractor, and plotted for times  $t \in [-0.15, 0.15]$ . For some segments, the  $C^0$  difference on the time interval  $[-0.15, 0.15]$  between the true and approximate solution is about  $r = 2 \times 10^{-13}$  while for others it is about  $r = 3 \times 10^{-8}$ .

- (a) the sequence of Taylor coefficients describes the function using only a countable collection of numbers, and
- (b) after fixing an appropriate choice of weights, the Taylor coefficients of an analytic function exhibit “rapid decay”.

In particular, these remarks suggest that analytic functions are well approximated by finitely many complex numbers, namely the classical Taylor polynomial approximation. In this chapter we are concerned first with approximating solutions of initial value problems by a finite Taylor polynomial, and second with bounding the truncation error associated with this approximation. Since we are interested in implementing this approximation on the computer it makes sense to frame the entire discussion in sequence space. This leads to a very natural infinite dimensional analogue of the algebraic nonlinear equations from Chapter 2. In order to formalize this notion we define a little more notation.

Let  $r > 0$  and  $z_0 \in \mathbb{C}$ . Denote  $D_r(z_0) = \{z \in \mathbb{C} \mid |z - z_0| < r\}$ . We set

$$C^\omega(D_r(z_0), \mathbb{C}) \stackrel{\text{def}}{=} \{f: D_r(z_0) \rightarrow \mathbb{C} \mid f \text{ is bounded and analytic}\}.$$

With sup norm,

$$\|f\|_\infty \stackrel{\text{def}}{=} \sup \{|f(z)| \mid z \in D_r(z_0)\},$$

$C^\omega(D_r(z_0), \mathbb{C})$  is a Banach space.

We want to understand the relationship between the Taylor coefficients of  $f$  and  $\|f\|_\infty$ .

We recall that in Example 7.3.3 we introduced the space  $\ell_{\nu, \mathbb{N}}^1$ , which is well suited to describe Taylor coefficients. For sake of simplicity of the notation in this Chapter, we denote  $\ell_\nu^1 = \ell_{\nu, \mathbb{N}}^1$ , and it should be clear from the context of Taylor series that the sequences of Taylor coefficients are always one-sided. Moreover, denote  $\|a\|_{1, \nu} = \|a\|_{1, \omega, \mathbb{N}}$  where  $\omega = \{\omega_k\}_{k \geq 0} = \{\nu^k\}_{k \geq 0}$ .

**Lemma 8.4.1.** *Let  $f \in C^\omega(D_r(z_0), \mathbb{C})$  with Taylor series given by  $\sum_{k=0}^\infty a_k(z - z_0)^k$ . If  $0 < \nu < r$ , then  $a = \{a_k\}_{k=0}^\infty \in \ell_\nu^1$ .*

*Proof.* Fix  $\nu \in (0, r)$  and  $r_* \in (\nu, r)$ . Since  $\overline{D_{r_*}(z_0)} \subset D_r(z_0)$ , then using the Cauchy Estimates (Theorem 19.5.4) we get that

$$|a_k| \leq \frac{1}{r_*^k} \sup_{z \in D_{r_*}(z_0)} |f(z)| = \frac{1}{r_*^k} \|f\|_\infty, \quad \text{for all } k \geq 0.$$

Hence,

$$\|a\|_{1, \nu} = \sum_{k=0}^\infty |a_k| \nu^k = \sum_{k=0}^\infty |a_k| r_*^k \left(\frac{\nu}{r_*}\right)^k \leq \sum_{k=0}^\infty \|f\|_\infty \left(\frac{\nu}{r_*}\right)^k = \frac{r_*}{r_* - \nu} \|f\|_\infty < \infty. \quad \square$$

Lemma 8.4.1 allows us to make the following definition.

**Definition 8.4.2.** The *Taylor transform* is the linear operator  $\mathcal{T}_{z_0, r, \nu}: C^\omega(D_r(z_0), \mathbb{C}) \rightarrow \ell_\nu^1$ ,  $0 < \nu < r$  that maps a bounded analytic function to the Taylor coefficients of its power series expansion at  $z_0$ .

**Theorem 8.4.3.** *The Taylor transform  $\mathcal{T}_{z_0, r, \nu}: C^\omega(D_r(z_0), \mathbb{C}) \rightarrow \ell_\nu^1$  is an injective bounded linear operator.*

*Proof.* Lemma 8.4.1 indicates that  $\|\mathcal{T}_{z_0, r, \nu}\|_{B(C^\omega(D_r(z_0), \mathbb{C}), \ell_\nu^1)} \leq \frac{r_*}{r_* - \nu}$  for any  $r_* \in (\nu, r)$ , and therefore

$$\|\mathcal{T}_{z_0, r, \nu}\| \leq \inf_{r_* \in (\nu, r)} \frac{r_*}{r_* - \nu} = \frac{r}{r - \nu}.$$

Since an analytic function is uniquely determined by its Taylor series coefficients,  $\mathcal{T}_{z_0, r, \nu}$  is injective.  $\square$

We can also work in the opposite direction.

**Lemma 8.4.4.** *Let  $a = \{a_k\}_{k=0}^\infty \in \ell_\nu^1$  and define*

$$f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k. \quad (8.46)$$

*Then  $f \in C^\omega(D_\nu(z_0), \mathbb{C})$  and  $\|f\|_\infty \leq \|a\|_{1,\nu}$ .*

*Proof.* The estimate

$$\begin{aligned} \|a\|_{1,\nu} &= \sum_{k=0}^{\infty} |a_k| \nu^k \\ &\geq \sup_{|z-z_0| < \nu} \sum_{k=0}^{\infty} |a_k| |z - z_0|^k \\ &\geq \sup_{|z-z_0| < \nu} \left| \sum_{k=0}^{\infty} a_k (z - z_0)^k \right| \\ &= \sup_{|z-z_0| < \nu} |f(z)| \\ &= \|f\|_\infty \end{aligned}$$

shows that the sum in (8.46) converges and hence that  $f \in C^\omega(D_\nu(z_0), \mathbb{C})$ .  $\square$

**Definition 8.4.5.** The *inverse Taylor transform* is the linear operator  $\mathcal{T}_{z_0,r,\nu}^{-1}: \ell_\nu^1 \rightarrow C^\omega(D_r(z_0), \mathbb{C})$ ,  $0 < r \leq \nu$  given by

$$\mathcal{T}_{z_0,r,\nu}^{-1}(a) = \sum_{k=0}^{\infty} a_k (z - z_0)^k.$$

Lemma 8.4.4 leads to the following theorem.

**Theorem 8.4.6.** *The inverse Taylor transform  $\mathcal{T}_{z_0,r,\nu}^{-1}: \ell_\nu^1 \rightarrow C^\omega(D_r(z_0), \mathbb{C})$  is an injective bounded linear operator.*

It is important to note that  $\mathcal{T}_{z_0,r,\nu}^{-1}$  is not the inverse of  $\mathcal{T}_{z_0,r,\nu}$  since for  $\mathcal{T}_{z_0,r,\nu}$  to be well-defined we need  $\nu < r$ , whereas for  $\mathcal{T}_{z_0,r,\nu}^{-1}$  to be well-defined we need  $r \leq \nu$ . Nevertheless, for  $\nu' < r \leq \nu$  and all  $a \in \ell_\nu^1$  we have

$$(\mathcal{T}_{z_0,r,\nu'}(\mathcal{T}_{z_0,r,\nu}^{-1}(a)))_k = a_k \quad \text{for all } k \in \mathbb{N},$$

so that  $\mathcal{T}_{z_0,r,\nu'}$  acts as the inverse on the range of  $\mathcal{T}_{z_0,r,\nu}^{-1}$ . Similarly, for  $r' \leq \nu < r$  and all  $f \in C^\omega(D_r(z_0), \mathbb{C})$  we have

$$\mathcal{T}_{z_0,r',\nu}^{-1}(\mathcal{T}_{z_0,r,\nu'}(f))(z) = f(z) \quad \text{for all } z \in \overline{B_{r'}(z_0)}.$$



Hence  $\mathcal{T}_{z_0, r', \nu}^{-1}(\mathcal{T}_{z_0, r, \nu'}(f))(z)$  can be analytically extended to  $D_r(z_0)$ , thus recovering  $f(z)$ .

We conclude that at the level of manipulating convergent Taylor series,  $\mathcal{T}$  and  $\mathcal{T}^{-1}$  do act as inverse operations, and we will drop the subscripts from the notation whenever convenient (i.e. always).

Since we want to deal with nonlinear dynamical systems it is important to understand how the Taylor transform interacts with differentiation and multiplication. We start with the latter, while the former will be discussed later in this section. The following lemma, which follows directly from Theorem 7.4.4, introduces the fundamental nonlinear operator on  $\ell_\nu^1$ .

**Lemma 8.4.7.** *Let  $a, b \in \ell_\nu^1$  and define the Cauchy product  $a * b$  by*

$$(a * b)_k = \sum_{j=0}^k a_{k-j} b_j, \quad (8.47)$$

for all  $k \geq 0$ . Then

$$\|a * b\|_{1, \nu} \leq \|a\|_{1, \nu} \|b\|_{1, \nu}.$$

In particular,  $*$ :  $\ell_\nu^1 \times \ell_\nu^1 \rightarrow \ell_\nu^1$  is a well defined binary operation which endows  $\ell_\nu^1$  with a Banach algebra structure.

Let  $a, b \in \ell_\nu^1$ . It follows from Lemma 8.4.7, and the discussion of Mertens' Theorem 7.4.3 that

$$\mathcal{T}^{-1}(a * b) = \mathcal{T}^{-1}(a) \cdot \mathcal{T}^{-1}(b), \quad (8.48)$$

which may also be written as

$$\mathcal{T}(\mathcal{T}^{-1}(a) \cdot \mathcal{T}^{-1}(b)) = a * b. \quad (8.49)$$

Since systems of differential equations are our main objects of study, the operator of differentiation plays a key role in the remainder of the book. Its action on  $C^\omega(D_r(z_0), \mathbb{C})$  and  $\ell_\nu^1$  is slightly subtle as the next example illustrates.

**Example 8.4.8.** Let  $z_0 = 0 \in \mathbb{C}$  and consider the sequence  $a \in \ell_1^1$  given by  $a_0 = 0$  and  $a_k = \frac{1}{k^2}$  for  $k \geq 1$ . The function  $\mathcal{T}^{-1}(a)$  given by  $f(z) = \sum_{k=1}^{\infty} \frac{1}{k^2} z^k$ , is analytic and bounded on  $D_1(0) \subset \mathbb{C}$  as  $\|f\|_\infty \leq \frac{\pi^2}{6}$ . The function  $f'$  has power series  $f'(z) = \sum_{k=1}^{\infty} \frac{1}{k} z^{k-1}$  which is analytic on  $D_1(0)$  but is no longer bounded. To see, observe that  $f'(z) = -\frac{\ln(1-z)}{z}$  for  $|z| < 1$ . In terms of sequence spaces we see that  $\mathcal{T}(f') = \{\frac{1}{k}\}_{k=1}^{\infty}$ , which is not in  $\ell_1^1$ . The discussion shows that differentiation is not a bounded linear operator on  $C^\omega(D_1(0), \mathbb{C})$  and its induced action on  $\ell_1^1$  is not bounded either. This readily generalizes to  $C^\omega(D_r(z_0), \mathbb{C})$  and  $\ell_\nu^1$  for any  $z_0 \in \mathbb{C}$  and any  $r, \nu > 0$ .

The following Lemma shows that even though differentiation is not well behaved as an operator from  $C^\omega(D_\nu(z_0), \mathbb{C})$  into itself, we can recover derivatives by “giving up” some portion of the domain of analyticity. Or, more pertinent to the viewpoint taking in this book, differentiation induces a bounded linear map from  $\ell_\nu^1$  into  $\ell_{\nu'}^1$  for any  $0 < \nu' < \nu$ . Moreover, we obtain an explicit bound on the norm of the operator.

**Lemma 8.4.9 (Cauchy Bounds for Analytic Functions).** *Let  $\nu > 0$ ,  $z_0 \in \mathbb{C}$  and  $a = \{a_k\}_{k \in \mathbb{N}} \in \ell_\nu^1$ . Define  $f: D_\nu(z_0) \rightarrow \mathbb{C}$  by  $f(z) \stackrel{\text{def}}{=} \sum_{k=0}^{\infty} a_k(z - z_0)^k$ . Let  $0 < \sigma \leq 1$  and  $\nu' = \nu e^{-\sigma} < \nu$ . Then  $\sup_{z \in D_{\nu'}(z_0)} |f'(z)| \leq \frac{1}{\nu\sigma} \|a\|_{1,\nu}$ .*

*Proof.* Since  $f$  is analytic in  $D_\nu(z_0)$

$$f'(z) = \sum_{k=1}^{\infty} k a_k (z - z_0)^{k-1},$$

for all  $z \in D_\nu(z_0)$ . Choose  $0 < \sigma \leq 1$  and define  $\nu' = \nu e^{-\sigma}$ . We leave it to the reader to check that  $k e^{-\sigma(k-1)} < 1/\sigma$  for all  $k \geq 0$ . Thus, for any  $z \in D_{\nu'}(z_0)$  we have that

$$|f'(z)| \leq \sum_{k=1}^{\infty} k |a_k| |z - z_0|^{k-1} \leq \sum_{k=1}^{\infty} k |a_k| (\nu e^{-\sigma})^{k-1} \leq \sum_{k=1}^{\infty} \frac{1}{\nu} k e^{-\sigma(k-1)} |a_k| \nu^k \leq \frac{1}{\nu\sigma} \|a\|_{1,\nu}. \quad \square$$

Let  $z_0 \in \mathbb{C}$  and  $0 < \nu' < \nu$ . Define the linear operator  $\mathcal{L}: \ell_\nu^1 \rightarrow \ell_{\nu'}^1$  by

$$\mathcal{L}(a) = \mathcal{T} \left( \frac{d}{dz} (\mathcal{T}^{-1}(a)) \right). \quad (8.50)$$

Lemma 8.4.9 shows that  $\mathcal{L}: \ell_\nu^1 \rightarrow \ell_{\nu'}^1$  is a bounded linear operator.

*Remark 8.4.10* (The differential as an unbounded operator). In a functional analytic setting it is often desirable to interpret the differential operator, or more generally an operator  $T$ , as mapping a Banach space  $\mathcal{X}$  to itself, even though that means the operator is not well-defined on all of  $\mathcal{X}$ . For example, this occurs when studying the eigenvalue problem

$$Tx = \lambda x \quad \text{with } \lambda \in \mathbb{C}.$$

Rather than expanding the range of the operator, one usually interprets  $T$  as having a restricted domain. In this context one usually speaks of *densely defined unbounded operators*. For example one can show that the set  $\mathcal{D} \subset \ell_\nu^1$  given by

$$\mathcal{D} = \left\{ a \in \ell_\nu^1 : \sum_{k=1}^{\infty} k |a_k| \nu^k < \infty \right\},$$

is dense in  $\ell_\nu^1$  and that the operator  $\mathcal{L}: \mathcal{D} \rightarrow \ell_{\nu'}^1$  defined by Equation (8.50) is well defined.

While this is the most convenient set up for spectral theory, it is not the approach taking in this book, as Theorem 7.6.2 applies to functions defined between different Banach spaces and Theorem 7.6.2 is the main tool for a-posteriori analysis used in all the applications we consider. We mention the theory of unbounded operators for the sake of completeness, and so that the reader is not confused when we do *not* adopt these conventions.

## 8.5 Exercises

**Exercise 8.5.1.** Let  $z_0 \in \mathbb{C}$  and  $0 < r' < r$ . Show that the differential operator  $f \rightarrow \frac{df}{dz}$  is a bounded linear operator from  $C^\infty(D_r(z_0), \mathbb{C})$  to  $C^\infty(D_{r'}(z_0), \mathbb{C})$ . Derive a bound on its operator norm.

**Exercise 8.5.2.** Consider a power series

$$P(t) = \sum_{k=0}^{\infty} p_k t^k,$$

and second power series

$$Q(t) = \sum_{k=0}^{\infty} q_k t^k.$$

Think of the power series coefficients of  $P(t)$  as “known”, and work out the recursion relations for the power series coefficients of  $Q(t)$ , in terms of the coefficients of  $P(t)$ , when

•

$$Q(t) = \frac{1}{P(t)}.$$

•

$$Q(t) = P(t)^{10}.$$

•

$$Q(t) = P(t)^\alpha.$$

•

$$Q(t) = \log(P(t)).$$

- $Q(t) = \sin(P(t))$ . Hint: work out the Taylor-coefficients of  $Q(t)$ , and  $R(t) = \cos(P(t))$  at the same time.
- For any complex number  $\alpha \in \mathbb{C}$  the Bessel function of order  $\alpha$  is the function  $B(x)$  solving the differential equation

$$x^2 \frac{d^2}{dx^2} B(x) + x \frac{d}{dx} B(x) + (x^2 - \alpha^2) B(x) = 0.$$

Compute the Taylor coefficient of the composition

$$Q(t) = B(P(t)).$$

- Let

$$P(t) = \frac{1}{1-t} = \sum_{k=0}^{\infty} t^k,$$

i.e. consider the power series coefficients  $p_k = 1$  for all  $n \geq 0$ . Write a computer program, which uses the recursion relations you computed above to compute the first 100 Taylor coefficients of

$$\left(\frac{1}{1-t}\right)^{10} = \sum_{k=0}^{\infty} q_k t^k.$$

Check the accuracy of your approximation by evaluating both sides of the above equation at  $t = -0.7, -0.5, -0.25, 0.25, 0.5$ , and  $0.75$ . Check the accuracy also at  $z = 0.5 + i0.65$ . Check the accuracy by plotting both sides of the above equation.

- Repeat the previous problem for

$$\sin(P(t)) = \sum_{k=0}^{\infty} q_k t^k,$$

$$\log(P(t)) = \sum_{k=0}^{\infty} q_k t^k,$$

and

$$P(t) = \frac{1}{(1-t)^{3/2}}.$$

**Exercise 8.5.3.** For this problem, repeat the work for the Lorenz system, changing the vector field  $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$  given by

$$f(x, y, z) = \begin{pmatrix} \alpha(y - x) + \delta xz \\ \rho x + \zeta y - xz \\ \beta z + xy - \epsilon x^2 \end{pmatrix},$$

with  $\alpha = 40$ ,  $\beta = 1.833$ ,  $\delta = 0.16$ ,  $\epsilon = 0.65$ ,  $\rho = 55$ , and  $\zeta = 20$ . More precisely:

- Derive the recursion relations for the power series coefficients of the solution.
- Write a computer program which solves these recursion relations to any desired order. Pick an initial condition and compute the Taylor expansion for the solution of the initial value problem. Plot the results. Try computing using various polynomial orders  $N$  and evaluating the polynomial on domain intervals of various lengths. Try to decide when the results look good and when you don't trust them.
- Write a program which solves the initial value problem and plots the results for a grid of initial points. Try to pick approximation order  $N$  and domain intervals so that all the curves "look good".

- “Evolve” your initial data points using either your new Taylor integrator or MatLab’s Runge-Kutta integrator until they are more or less “on the attractor”. Compute and plot the Taylor expansions for this new data set.

(Note that my solution of this problem for the Lorenz system is posted online. You can use my codes as a jumping off point).

**Exercise 8.5.4.** Same as the previous problem, but for the vector field  $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$  given by

$$f(x, y, z) = \begin{pmatrix} (z - \beta)x - \delta y \\ \delta x + (z - \beta)y \\ \gamma + \alpha z - \frac{z^3}{3} - (x^2 + y^2)(1 + \epsilon z) + \zeta z x^3 \end{pmatrix}.$$

Take  $\epsilon = 0.25$ ,  $\alpha = 0.95$ ,  $\gamma = 0.6$ ,  $\delta = 3.5$ ,  $\beta = 0.7$  and  $\zeta = 0.1$ . This is known as the Aizawa System. For the cubic term you can use the formula

$$\left( \sum_{k=0}^{\infty} a_k t^k \right)^3 = \sum_{k=0}^{\infty} (a * a * a)_k t^k,$$

and also that

$$\left( \sum_{k=0}^{\infty} a_k t^k \right)^3 \left( \sum_{k=0}^{\infty} b_k t^k \right) = \sum_{k=0}^{\infty} (a * a * a * b)_k t^k.$$

Alternatively, you could use automatic differentiation to handle the cubic term.

**Exercise 8.5.5.** Same as the previous two problems, but for the vector field  $f: \mathbb{R}^3 \rightarrow \mathbb{R}^3$  given by

$$f(x, y, z) = \begin{pmatrix} \alpha(y - x) + \delta x z \\ \rho x + \zeta y - x z \\ \beta z + x y - \epsilon x e^{-x} \end{pmatrix},$$

with  $\alpha = 40$ ,  $\beta = 1.833$ ,  $\delta = 0.16$ ,  $\epsilon = 0.65$ ,  $\rho = 55$ , and  $\zeta = 20$ . So you need to use automatic differentiation on the exponential function.

**Exercise 8.5.6.** Use automatic differentiation to repeat the previous problems but for the pendulum.

**Exercise 8.5.7.** Consider the system of Exercise 8.5.3.

- Choose the initial condition  $x_0 = 1.5$ ,  $y_0 = -2$ ,  $z_0 = 2$ , and compute the Taylor coefficients of the solution of the initial value problem to order  $N = 50$ . Now implement the bounds and derive the appropriate radii polynomial, using  $\nu = 0.5$ . Check that you obtain computer assisted error bounds.
- Repeat the process above for different values of  $\nu$  and  $N$ . Validate the longest orbit segment you can.

- Repeat at other initial conditions.

**Exercise 8.5.8.** Consider the Aizawa system of Exercise 8.5.4.

- Choose the initial condition  $x_0 = 1.5$ ,  $y_0 = -2$ ,  $z_0 = 2$ , and compute the Taylor coefficients of the solution of the initial value problem to order  $N = 50$ . Now implement the bounds and derive the appropriate radii polynomial, using  $\nu = 0.5$ . Check that you obtain computer assisted error bounds.
- Repeat the process above for different values of  $\nu$  and  $N$ . Validate the longest orbit segment you can.
- Repeat at other initial conditions.

**Exercise 8.5.9.** Consider the system of Exercise 8.5.5.

- In order to manage the exponential nonlinearity, append an appropriate equation (i.e. use “automatic differentiation” to transform the equation to an equivalent polynomial system).
- Choose the initial condition  $x_0 = 1.5$ ,  $y_0 = -2$ ,  $z_0 = 2$ , and compute the Taylor coefficients of the solution of the initial value problem to order  $N = 50$ . Now implement the bounds and derive the appropriate radii polynomial, using  $\nu = 0.5$ . Check that you obtain computer assisted error bounds.
- Repeat the process above for different values of  $\nu$  and  $N$ . Validate the longest orbit segment you can.
- Repeat at other initial conditions.

**Exercise 8.5.10.** Same problem as above, but for the pendulum.

**Exercise 8.5.11.** Same problem as above, but for the Kepler problem.